

Fast and Slow Arbitrage: The Predictive Power of (Persistent) Capital Flows for Factor Returns

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Abstract

We document that persistent aggregate capital flows to hedge and mutual funds predict monthly factor returns with an out-of-sample R^2 reaching 6.6%. Transient flows display no such power despite being more predictable. We show—both empirically and theoretically—that persistent flows’ distinctive predictive power stems from active fund managers’ capital constraints. As a result, managers invest persistent, but not transient, capital flows into factor trading strategies, leading to factor return predictability and factor momentum, yet greater price efficiency. Our key insight is that capital-constrained managers account for both current and anticipated future capital flows in the arbitrage sector, thereby incorporating the dynamics of capital into their strategies.

Keywords: capital management; factor return predictability; capital flows; anomalies; factor momentum; hedge funds; mutual funds; slow-moving capital; leverage constraints; limits to arbitrage; out-of-sample and in-sample predictability; spectral analysis.

Introduction

Recent studies have revealed that anomaly returns (a.k.a. factors) are predictable. This finding has far-reaching implications for asset pricing: not only does factor return predictability call into question market efficiency in its weakest form, but it is also essential for understanding the shape of the stochastic discount factor (SDF) at the heart of every asset pricing model (see, e.g., the classic Hansen and Jagannathan (1991) bound).¹ In this paper, we investigate whether, and how, the dynamics of capital channeled to financial intermediaries—specifically arbitrageurs—affect the predictability of factor returns. Indeed, a growing body of research shows that time variations in financial intermediaries' capital and in their constraints determine the conditional properties of asset returns.

We uncover a powerful predictor of factor returns:² the *persistent* components of aggregate capital flows into the arbitrage sector, with the hedge fund sector (and to a lesser extent the mutual fund sector) serving as the most relevant proxy. These flows predict common factor returns with far greater power (based on their out-of-sample (OOS) R^2) than previously reported predictors.³ Remarkably, total flows into hedge funds and their transient components display no such OOS predictive power—even though transient flows are more predictable than persistent flows. We show—both empirically and theoretically—that persistent flows' distinctive predictive power stems primarily from active fund managers' capital constraints. Our key insight is that capital-constrained managers, in making investment decisions, not simply consider the supply of capital currently available in the arbitrage sector, but instead consider the current supply relative to the anticipated future supply of capital. In other words, they incorporate the dynamics of capital into their strategies. As a result, managers invest persistent, but not transient, capital flows into factor trading strategies, thereby generating factor-level predictability.

¹ Factor return predictability implies that a dynamic combination of factors outperforms static positions in these factors, implying a SDF with higher volatility, heteroskedasticity, and price of risk than generated by leading asset pricing models (Haddad et al. 2020; Kelly et al. 2021).

² Unlike those used in the literatures on market or individual stock return predictability, predictors of (cross-sectional) factor returns have only recently been discovered. Those broadly fall into two categories. 1) Those that exploit *factor characteristics* (e.g., long-minus-short valuation ratios). For specific anomalies, see, e.g., Cohen, Polk, and Vuolteenaho (2003), Greenwood and Hanson (2012), Kelly and Pruitt (2013), and Baba Yara, Boons, and Tamoni (2021). Haddad et al. (2020) and Kelly et al. (2021) analyze the component common to many anomalies. 2) Some predictors are based on *lagged factor returns* (e.g., factor momentum). See, e.g., McLean and Pontiff (2016), Avramov et al. (2017), Arnott et al. (2019), Gupta and Kelly (2019), Alti and Titman (2019), Falck, Rej, and Thesmar (2020), and Ehsani and Linnainmaa (2022a).

³ This R^2 determines how factor return predictability shapes the SDF and the maximum Sharpe ratio. Indeed, Haddad et al. (2020) establish a general theoretical linkage whereby the R^2 of the predictive regression determines the maximum Sharpe ratio, which in turn determines the minimum variance of the SDF (see Equation 7 in Haddad et al. 2020).

Perhaps counterintuitively, we show theoretically that this predictability is associated with more efficient prices (i.e., closer to fundamentals). In contrast, transient flows have little impact on the dynamics of factor returns.

To come to these conclusions, we extract the slow- and fast-moving (a.k.a, persistent and transient) components of monthly flows and factor returns using standard and easily implementable filters, namely a 6-month moving average and monthly changes, respectively. The anomalies we focus on are the anomaly concepts examined in Stambaugh, Yu, and Yuan (2012, 2015)—denoted “SYY”—whose returns are shown to be closely related to hedge- and mutual-fund flows by Akbas et al. (2015).

We show that the persistent component of hedge-fund flows predicts factor returns with an OOS R^2 of 6.6% over the period 1994–2020. To put this number into perspective, Campbell and Thompson (2007) view as economically significant an OOS R^2 of 0.5% for predicting the aggregate market return, while Haddad et al. (2020) report an OOS R^2 around 4% for factor timing portfolios. Moreover, persistent hedge-fund flows have far more OOS predictive power than extant factor return predictors and subsume most of them in encompassing tests. Remarkably, total flows and their transient component have *no* predictive power. Persistent, but not transient, mutual-fund flows also possess predictive power (albeit much weaker than hedge funds’), although they are a less accurate proxy for arbitrage capital.

Why do persistent—but not transient—flows predict factor returns? To answer this question without being confined to specific filters for estimating those components, we analyze the series in the frequency domain. Spectral analysis allows—without imposing any structure on the data—to decompose any (stationary) series into orthogonal components, each associated with a frequency measuring how many cycles are completed per unit of time. For example, a one-year cycle corresponds to a six-month half-life. Lower-frequency components are slower moving, i.e., more persistent or less transient.⁴ To be clear, our notion of frequency focuses on the dynamics of a series, i.e., its degree of persistence, and is distinct from the frequencies of rebalancing (i.e., turnover) and sampling, and from the holding horizon. We split the flow and return series into a “low-frequency” component corresponding to frequencies of one cycle per year or less, and a “high-frequency” component corresponding to frequencies of more than one cycle per year—to which we refer interchangeably as persistent

⁴ Our description of low-frequency flows as “slow” can be related to the half-life of a process, i.e., the time it takes the process to lose half of its previous value. Suppose flows experience a positive shock. If they comprise more low-frequency components, flows will need more time to return to their mean. As a result, they will remain elevated for an extended period, i.e., their half-life is longer.

and transient, respectively. This orthogonal decomposition makes it possible to simply and accurately evaluate each component's contribution to properties of the original series, including to return predictability. Notwithstanding, our main findings are robust to using a variety of filters.

In this framework, the R^2 from predicting returns with flows at a given frequency is given by the product of two coefficients estimated over that frequency: i) the (squared) *serial correlation of flows*, which reflects how predictable flows are at that frequency; and ii) *the R^2 of a contemporaneous regression of returns on flows* over that frequency, which represents how closely flows and returns are related contemporaneously. Regarding i), we find that flows' moving average and monthly changes are both highly predictable, with OOS R^2 's around, respectively, 5% and 17% for hedge funds. Thus, the inability of changes in flows to predict returns is not caused by those flows being themselves unpredictable—in fact, they are even more predictable OOS than flows' moving average. This puts the onus on the contemporaneous link between flows and returns to explain differences between flows' moving average and monthly changes in their predictive power for factor returns. Accordingly, we analyze, over each frequency, how flows and returns are related contemporaneously.

We find that the R^2 of the contemporaneous regression of returns on hedge-fund flows is significantly larger at low frequency than at high frequency. Thus, persistent flows explain a considerable proportion of persistent factor return variations—in fact, their explanatory power far exceeds that of any factor from recent prominent models. In contrast, transient flows have virtually no explanatory power for transient factor returns. With persistent factor returns accounting for close to a quarter of variations in total factor return, this *tilt in explanatory power towards low frequency* is the reason that only persistent flows predict factor returns.

By definition, differences in R^2 across frequencies must be driven by differences in the volatility of flows (the regressor) and/or in the regression coefficient. For hedge funds, which are our primary focus, the volatility of flows is similar across frequencies, whereas the regression coefficient is multiple times larger at low frequency than at high frequency. Thus, their low-frequency tilt stems entirely from differences in the regression coefficient. Their tilt is far stronger than mutual funds' tilt.

Regarding the regression coefficients, we argue—based on our theory (described below) and prior evidence that hedge funds are arbitrageurs—that they should be positive for hedge funds at low frequency in both contemporaneous and predictive regressions, reflecting their role in correcting mispricing. For mutual funds in contrast, coefficients should be negative in contemporaneous regressions, given prior evidence indicating that

mutual funds exacerbate pricing errors, and turn positive in predictive regressions, as their errors get corrected, e.g., by hedge funds. Our tests validate these expectations.

In the remainder of the analysis, we further investigate the economic mechanisms behind the low-frequency tilt and the ensuing ability of persistent, but not transient, hedge-fund flows to predict factor returns. The contemporaneous regression coefficient, which drives the tilt, reflects how fund *managers* deploy the capital they receive since it measures how a unit of flows impacts factor returns. Our finding that lower frequency flows have a *bigger* effect on factor returns may therefore seem surprising. Indeed, deploying flows faster into stocks—as one would expect managers do with higher-frequency flows—should trigger a larger price impact because of more limited liquidity provision over shorter horizons. Therefore, our finding suggests that managers do not passively pass on flows to the stock market but rather actively attenuate the effect of higher-frequency flows; in other words, they aggressively invest persistent capital on factor strategies, while transient capital is repurposed. This behavior contrasts sharply with passive fund managers' who have no discretion in dealing with flows; and indeed, we find that flows to passive funds have a stronger effect at *higher* frequency on the market factor and no effect on cross-sectional factors.

We make sense of the low frequency tilt and the role of managers therein in a three-period model featuring rational constrained fund managers and exogenous factors—which, for ease of exposition, are interpreted as mispricing but which could be viewed as compensation for risk. Managers, a.k.a., arbitrageurs, know that the mispricing will eventually disappear (for exogenous reasons), and so they trade on (that is, correct) it now. But because arbitrage capital is limited, the extent to which they do so depends on the supply of arbitrage capital and its dynamics. Suppose there is an inflow of capital in period 1. If flows are persistent, then arbitrageurs anticipate more inflows in the *next* period, and therefore, large mispricing corrections. This leads them to trade aggressively on the mispricing immediately. But because capital is limited in period 1, they cannot drive expected returns to zero, which results in factor return predictability. In this case, prices are closer to fundamentals than when flows are transient. If instead flows are transient, then managers anticipate a moderate price correction in period 2 since arbitrage capital will then be scarce. In this case, the capital in period 1 suffices to drive the expected return to zero, rendering returns unpredictable. Thus, what determines return predictability is not the current supply of capital—it is the same in both cases—but its current supply *relative* to its expected future supply.

The model generates several predictions. First, persistent, but not transient, flows predict returns. Second,

they have more explanatory power for returns in contemporaneous regressions and yield a larger regression coefficient than do transient flows. As discussed, both predictions conform to the data. The model also yields three ancillary predictions which illuminate the mechanism at work and its implications. First, arbitrageurs trade the mispriced asset more aggressively when flows are more persistent. We document that managers indeed take larger net positions in anomaly strategies and in illiquid stocks when they receive lower-frequency flows. Second, (persistent) flows' predictive power for returns hinges on arbitrage capital being scarce. Exploiting variations in funding costs over time and in leverage capacity in the cross-section, we report that it indeed weakens when capital becomes more abundant. Furthermore, mispricing is slow to correct only for stocks traded exclusively by constrained funds, so only those stocks should exhibit predictability. This implication is confirmed in the data. The final prediction of the model is that persistent flows—a.k.a. flow momentum—generate factor momentum. Extracting from factor returns components associated with persistent and transient flows, we document that the former drives factor momentum.

Finally, while the collection of our results—both empirical and theoretical—supports a causal interpretation of the effect of (persistent) flows on factor returns, one may still be concerned about an omitted variable or reverse causality. To further alleviate those concerns, we conduct tests that exploit quasi-exogenous variations in flows. Our main test uses the “Granular Instrumental Variable” (GIV) approach of Gabaix and Koijen (henceforth GK) (2022a,b) to extract, from fund flows, shocks orthogonal to economy-wide or fund sector-wide variables that could influence factors. Those tests confirm the presence of the low-frequency tilt.

Related literature and contribution

Our work contributes to three streams of research. First, we add to the literature on intermediary asset pricing and, more broadly, to that on the limits to arbitrage. This literature has shifted the focus from households to financial intermediaries, arguing that constraints on their capital govern the conditional properties of returns. We show that due to time variations in capital available to constrained arbitrageurs—specifically hedge and mutual funds—the persistent component of capital flows contains predictive information for factor returns. Our explanation for this phenomenon—supported by empirical and theoretical analyses—recognizes that fund managers base their investment strategies not only on current capital but also expected future capital in the

arbitrage sector.⁵ According to our mechanism, returns exhibit greater predictability when flows are more persistent despite prices being *closer* to fundamentals, underscoring the importance of understanding the implications of flow persistence for market efficiency. Extant papers in the intermediary asset pricing literature focus on capital constraints faced by broker-dealer types of intermediaries, which are more relevant to complex asset classes (He and Krishnamurthy 2012, 2013; Adrian, Etula, and Muir 2014; He, Kelly, and Manela 2017).⁶

Within that literature, our study also is related to, but distinct from, the notion of “slow-moving capital” (e.g., Duffie 2010). We differentiate, in the movement of capital, the provision of capital (by investors) from the processing of capital (by managers). Furthermore, while slow-moving capital is associated with price inefficiency in the literature, it is not explicitly linked to return predictability as in our model. Our findings suggest that arbitrageurs’ capital is not slowly supplied by investors, but rather selectively deployed by managers. Indeed, managers aggressively invest only slow-moving (i.e., persistent) capital, which leads to predictable returns and yet prices that are more efficient than those resulting from fast-moving (i.e., transient) capital.

Second, our work belongs to the recent body of research that conducts meta-analyses to understand factor return predictability and its implications (Haddad et al. 2020; Kelly et al. 2021).⁷ Our contribution to this body is to introduce a new category of predictor, along with an economic mechanism distinct from those proposed for existing predictors. This predictor—*persistent* flows (especially to hedge funds)—exhibits more predictive power OOS for the systematic component of an important class of anomalies compared to previously reported major predictors. Transient flows have no such power; neither do persistent flows to passive funds. Thus, our findings underscore that persistent flows to active funds contain information about expected factor returns. Importantly, our focus on OOS evidence sets us apart from most studies on factor predictability. Indeed, the implications for market efficiency and the SDF can be reliably drawn from OOS, but not in-sample, tests.⁸

⁵ Factor returns differ from stock returns in that they involve dynamically (e.g., monthly) rebalanced strategies. As a result, they are based on a different set of stocks each month. Thus, our findings speak to the effect of flow persistence on the evolution of factor-level returns rather than on that of mispricing in a single stock.

⁶ Kozak, Nagel, and Santosh (2018) propose a model of factor returns, with rational arbitrageurs trading against sentiment investors. Ehsani and Linnainmaa (2022a) extend this model to account for persistent sentiment. In this class of models, factor returns are related positively to sentiment and negatively to arbitrage capital. In contrast, in our model, factor returns are positively related to arbitrage capital flows (see Internet Appendix I9).

⁷ While earlier studies document predictability for a single anomaly (see Footnote 2), these two recent papers adopt a systematic (meta-analysis) approach by analyzing many anomalies at once and extracting their common components. This dimension-reduction makes it possible to estimate an SDF and avoid data mining (e.g., Haddad et al. 2020). Moreover, managers in practice typically trade on composite signals across anomalies rather than on any individual anomaly (See, e.g., Cho 2020). These two studies also focus on OOS tests.

⁸ There are three reasons. First and foremost, as Martin and Nagel (MN, 2022) argue, “modern investors face a high dimensional prediction problem,” whereby many variables may predict asset prices. Nowadays, the number of parameters governing a predictor’s process which need to be estimated can be close to or even larger than the number of factor return observations. In this case, “evidence of ... return predictability from in-sample regressions does not tell us much about the expected returns that investors perceived *ex ante* at the

Furthermore unlike studies in this literature, we address endogeneity concerns by exploiting granular instruments (GK 2022 a,b) and seasonal flow patterns (Etula, Rinne, Suominen, and Vaittinen 2020).

In addition, our findings have broader implications for the second moments of asset returns, namely their i) serial correlation, ii) volatility and iii) cross-correlations. i) The first relates to a central finding in the literature on factor returns that factors display pervasive momentum (see Ehsani and Linnainmaa 2022a,b and other references in our Footnote 2).⁹ ii) The second implication concerns demand-based asset pricing, particularly the relation between capital flows and stocks' excess volatility. GK (2022a) estimates that aggregate stock market fluctuations (i.e., the market factor) are primarily generated by flows in and out of the market. Extending this study, we examine flow-driven price fluctuations in cross-sectional factors and across frequencies. Furthermore, Kokkonen and Suominen (2015) and Akbas et al. (2015) report that (total) flows to hedge and mutual funds are *contemporaneously* associated with factor returns.¹⁰ We complement these studies by demonstrating, through OOS tests, that persistent, but not total nor transient, flows have strong predictive power. We also provide direct evidence tying together flows, managers' trades, and factor return fluctuations. iii) The final implication pertains to a broad range of assets' return predictability. Indeed, the mechanism underlying our findings—that persistent trading (of a potentially broad range of market participants) on mispricing generates return predictability—applies across many mispriced assets, across asset classes, and across time, thereby connecting their returns (e.g., from stocks to bonds in Pitkäjärvi, Suominen, and Vaittinen 2020; from micro-level to macro-level in Dong, Li, Rapach, and Zhou 2022; persistent retail trading in Dong and Yang 2023; across time in Da and Zhang 2024; and persistent firm share issuance and repurchase in Da, Dong, Wu, and Zhou 2025).

Our final contribution is to the nascent literature that breaks up financial variables of interest into their low-

time they priced assets,” and “Only out-of-sample tests can do so.” This realistic setting differs from traditional rational expectations models that specify a null hypothesis based on a population model with truly unpredictable returns and a known underlying process generating returns. Second, only OOS tests imply strategies that market participants can actually implement. Third, they are not subject to potential upward biases involving persistent predictors (see Section 4 in Boudoukh et al. 2021).

⁹ Lou (2012) reports that the tendency of mutual fund investors to chase performance generates stock-level momentum. Vayanos and Woolley (2013) argues that stock-level momentum can be caused by rational prices underreacting to expected future flows, Ben-David et al. (2021) show that Morningstar ratings drive mutual fund flows and, until their methodology was reformed in 2002, affected size and value style-portfolios. We differ from those studies along two important dimensions. First, most of these papers study the generation of stock-level returns, i.e., the unconditional existence of factor returns. In contrast, we analyze the (OOS) predictability of factor-level returns, which pertains to the conditional properties of factor returns (see also Footnote 5). We also examine multiple factors at once, aggregated into a single composite factor. As Haddad et al. (2020) show, such *factor-level predictability* is crucial for developing asset pricing models. Second, we emphasize the difference between persistent and transient flows whereas they make no such distinction. Critically, we find that only persistent flows affect factors and relate this effect to managers' (rather than investors') behavior. Third, we focus on hedge funds.

¹⁰ Coval and Stafford (2007), Edelen, Ince, and Kadlec (2016), Chen, Da, and Huang (2019), Calluzzo, Moneta, and Topaloglu (2019), Huang, Song, and Xiang, (2019), and Li (2022) also focus on the relation between institutional trading and stock/factor returns. We differ from these papers for one or more of the three reasons outlined in Footnote 9.

and high-frequency drivers. Models such as Dew-Becker and Giglio (2016), Bidder and Dew-Becker (2016) and Crouzet et al. (2020) work out the theoretical implications of frequency components for asset pricing. Empirical studies conduct frequency analyses of stock price-related and risk measures.¹¹

1. Data and variable construction

1.1 Anomalies

Our main dependent variable is the return on the composite factor, which we construct based on prominent anomaly concepts discussed in SY Y (2012, 2015).¹² As in Haddad et al. (2020), dimension reduction “allows for robust estimation of factor predictability, and therefore the SDF. (p. 1981)” Furthermore, SY Y (2015) shows that returns to individual anomalies have low correlations with each other but relatively high correlations with the returns to a long-short strategy that combines their anomalies, suggesting that each captures a different element of aggregate cross-sectional mispricing. They argue that this long-short strategy “diversifies away [the] noise in each individual anomaly and . . . increases precision” for measuring aggregate mispricing. Accordingly, we generate, for each anomaly set, a time-series of aggregate returns. We construct the composite anomaly portfolio by averaging each month signals across anomalies, and going long (short) the stocks that appear most undervalued (overvalued). For signals from quarterly (annual) statements, we follow the convention by lagging them by 4 (6) months to ensure that the investing public including fund managers can observe them in the ranking month. Importantly, our focus throughout is on the properties of anomaly *strategies*—which rebalance stock portfolios monthly—rather than on those of stocks. Our main results hold when employing principal component analysis (PCA) as an alternative method for dimension reduction (see Internet Appendix I8 Table I8-2).

Although we do not critically rely on whether anomalies are interpreted as risk or mispricing, we use the mispricing-based interpretation to simplify the exposition. Under this interpretation, the returns on the long-short strategy in the post-ranking month determine whether mispricing is corrected or exacerbated in that month. Specifically, stocks in the long leg are undervalued at the end of month t . Hence a positive return on the long leg

¹¹ Calvet and Fisher 2007; Yu 2012; Ortu, Tamoni, and Tebaldi 2013; Dew-Becker and Giglio 2016; Chinco and Ye 2017; Schneider 2019; Chaudhuri and Lo 2019; Neuhierl and Varneskov 2021; Bandi, Chaudhuri, Lo, and Tamoni 2021.

¹² SY Y anomaly concepts are the return on assets (Chen, Novy-Marx, and Zhang 2010), O-score (Ohlson 1980), failure probability (Campbell, Hilscher, and Szilagyi 2008), gross profitability (Novy-Marx 2013), net stock issuances (Ritter 1991; Loughran and Ritter 1995), accruals (Sloan 1996), momentum (Jegadeesh and Titman 1993), composite equity issuance (Daniel and Titman 2006), gross net operating assets (Hirshleifer et al. 2004), asset growth (Cooper, Gulen, and Schill 2008), and the ratio of investments to assets (Titman, Wei, and Xie 2004).

in month $t + 1$ indicates that these stocks have appreciated and are less undervalued. Analogously, stocks in the short leg are overvalued in month t , so a negative return on the short leg in month $t + 1$ indicates a correction of overvaluation. In months during which aggregate mispricing is corrected (exacerbated), the long leg realizes higher (lower) returns than the short leg, and so returns on the long-short strategy are positive (negative).¹³

1.2 Hedge and mutual funds: flows and trading

Monthly aggregate fund flows to hedge funds (HF) and actively managed mutual funds (MF) are computed as

$$HF_t \text{ (or MF}_t\text{)} = \frac{\sum_{i=1}^N TNA_{i,t} - TNA_{i,t-1}(1 + R_{i,t})}{\sum_{i=1}^N TNA_{i,t-1}},$$

where $TNA_{i,t}$ represents the total net assets of fund i in month t and $R_{i,t}$ is the return on fund i over month t . We obtain monthly total net assets and returns from the CRSP Survivor Bias-Free U.S. Mutual Fund and the Lipper TASS Hedge Fund databases. We follow the procedure of Huang, Sialm, and Zhang (2011) to select actively managed mutual funds primarily investing in U.S. equities. Likewise, we only include U.S. equity-focused hedge funds (both alive and dead). The fund data cover the period from January 1994 to December 2020.

We proxy for funds' trades using quarterly changes in their holdings from 13F filings. We identify hedge and mutual funds using the classifications of Agarwal et al (2013) and of Koijen and Yogo (2019), respectively.

1.3 Control variables and other data

Table 1: Summary Statistics

Panel A presents descriptive statistics for main variables. Panel B reports the serial correlations of filtered returns and flows. Panel C displays the variance decomposition of filtered returns and fund flows. Panel D provides the correlations between the variables filtered using MA(6) and change and those using spectral analysis. MF and HF are the monthly aggregate percentage flows to US-equity oriented active mutual funds and hedge funds, respectively. Passive is the monthly aggregate percentage flow of US-equity oriented passive mutual funds. SYY is the return of the long-minus-short strategy based on anomalies in SYY (2012). Amihud and Turnover denote the equally-weighted average of, respectively, the Amihud illiquidity ratio and turnover of all common stocks listed on NYSE in month t . MKTRF is the monthly return of the market in excess of the risk free rate. HML and SMB are the monthly returns to the value and the size strategies, respectively. We use MA(6), 6-monthly moving average, as a low-pass filter, and monthly change, the first difference, as a high-pass filter. The LOW (HIGH) components are time series that are re-constructed from frequencies that have cycles longer than or equal to (shorter than) one year. The sample period is 1994–2020.

Panel A: Descriptive Statistics

Variable (%)	N	Mean	t Value	Std Dev	Q1	Median	Q3
SYY	324	1.25	[3.72]	6.07	-1.20	1.67	4.53
HF	324	0.34	[3.89]	1.58	-0.35	0.45	1.27
(Active) MF	324	0.06	[2.05]	0.55	-0.29	0.00	0.41
Passive	324	0.86	[14.13]	1.10	0.17	0.69	1.38
MKTRF	324	0.73	[2.96]	4.47	-1.96	1.33	3.47
Amihud	324	3.44	[25.43]	2.43	1.47	2.54	5.14
Turnover	324	164.22	[41.21]	71.72	95.09	176.29	210.29
HML	324	0.02	[0.10]	3.14	-1.80	-0.04	1.61

¹³ Average returns on the long leg, short leg, and long-short strategy are, respectively, positive, negative and positive in our sample, consistent with the literature. Hence, the correction of mispricing tends to dominate its worsening.

SMB	324	0.12	[0.67]	3.25	-1.91	0.09	2.04
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Panel B: Auto Correlations of Filtered and Decomposed Variables

Filter \ Variables	SYY		HF		MF	
	Rho	t value	Rho	t value	Rho	t value
Total	0.131	[1.50]	0.445	[4.29]	0.628	[11.62]
LOW	0.955	[52.07]	0.967	[32.40]	0.979	[72.40]
HIGH	-0.103	[-1.85]	-0.126	[-1.97]	-0.054	[-1.28]
MA(6)	0.866	[27.80]	0.953	[30.67]	0.965	[59.46]
Change	-0.426	[-11.14]	-0.422	[-10.46]	-0.387	[-10.71]

Panel C: Variance Decomposition (Ratio to Total Variance)

Filter \ Variables	SYY	HF	MF
LOW	22.1%	52.2%	66.8%
HIGH	77.9%	47.8%	33.2%
MA(6)	21.1%	46.6%	60.8%
Change	174.0%	108.9%	68.7%

Panel D: Correlations between Filtered and Decomposed Variables

Variables	SYY		HF		MF	
	Correlation	p value	Correlation	p value	Correlation	p value
Corr(LOW, MA(6))	0.89	[0.00]	0.96	[0.00]	0.98	[0.00]
Corr(HIGH, Change)	0.74	[0.00]	0.74	[0.00]	0.69	[0.00]

The stock sample includes all common stocks listed on the NYSE, AMEX, and Nasdaq. We follow Akbas et al. (2015) and control for commonly used risk and aggregate liquidity factors. These include: *MKTRF*, the monthly market return in excess of the risk-free rate; *HML* and *SMB*, the monthly returns to value and size strategies; *Amihud*, the equal-weighted average of the Amihud (2002) illiquidity ratio of all common stocks listed on the NYSE in month t ; and *Turnover*, the equal-weighted average turnover of all common stocks listed on the NYSE in month t . Table 1 shows descriptive statistics of main variables and their characteristics.

2. Predictive power of flows for factor returns

2.1 Out-of-sample predictive power

Table 2: Out-of-sample (OOS) Forecasting Performance – Factor Returns on Fund Flows

The table reports out-of-sample R^2 (based on the methodology of Campbell and Thompson (2008)) of forecasting regressions of factor returns on filtered fund flows. We use 6-month moving averages (MA(6)) and first differences (Change) as low- and high-pass filters. For each hedge or mutual funds, we estimate the following regression:

$$HF_{t-1}^K (MF_{t-1}^K) = \gamma_0 + \gamma_1 r_{t-1} + \gamma_2' X_{t-1} + \varepsilon_{t-1}^{HF,K} (\varepsilon_{t-1}^{MF,K}),$$

where K = Raw (raw flows), MA(6), or Change, r_{t-1} is factor return in month $t-1$, and X_{t-1} is a set of control variables, including the Fama-French three factors, the aggregate turnover, and the aggregate Amihud illiquidity. Then, using the regression residuals, we estimate the following forecasting regression:

$$r_t = \alpha + \beta \varepsilon_{t-1}^{HF,K} (\varepsilon_{t-1}^{MF,K}) + \epsilon_t,$$

Finally, out-of-sample forecast for r_{t+1} is obtained from $\hat{\alpha} + \hat{\beta} \varepsilon_t^{HF,K} (\hat{\beta} \varepsilon_t^{MF,K})$. The full sample covers 1994 to 2020, and the initial in-sample estimation period is 1994–2003. We use the recursive estimation method. The dependent variables are the returns of the long-minus-short strategy based on anomalies in SYY (2012). The benchmark predictor in computing OOS R^2 is the recursive mean of the factor returns in the corresponding predictive regression. The lower bound of the confidence interval is calculated based on the

bootstrapped t -statistics of Clark and West (2007). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for R^2 statistics. Coefficients of interest that are statistically significant are bolded.

		(1)	(2)	(3)	(4)	(5)	(6)
Fund Type		Hedge Fund			Mutual Fund		
Flows		Raw	MA(6)	Change	Raw	MA(6)	Change
$R^2(\%)$	Point Estimate	0.19	6.59***	-3.71	2.82**	3.65***	-0.64
	Lower Bound (10%)	-0.04	3.20	-7.29	1.04	2.06	-1.50
	Lower Bound (5%)	-0.11	2.23	-8.31	0.54	1.61	-1.74

We start with predictive regressions of factor returns (in month t) on flows and their components (in month $t-1$). Flows' persistent and transient components are computed using 6-month moving averages (from $t-1$ to $t-6$) and monthly changes (from $t-1$ to $t-2$), respectively.¹⁴ We conduct a standard recursive OOS estimation with an expanding window: using data from the first n months of our sample ($n \geq 120$), we predict factor returns in the $n+1^{th}$ month. To control for sources of predictability not specific to flows such as factor momentum, we first estimate monthly regressions of flow variables on SYR returns and the five risk and liquidity factors employed in Akbas et al. (2015), and then use the residuals to predict the next-month factor return.¹⁵ We generate such residual flows within each in-sample estimation period to make sure that our test is strictly OOS. We report in Table 2 the OOS R^2 for three measures of flows (total, persistent and transient) for both hedge and mutual funds. Following Campbell and Thomson (2008), the benchmark predictor for computing OOS R^2 is the recursive mean factor return in the corresponding predictive regression. We establish a lower bound for the confidence interval for R^2 using the bootstrapping methodology of Clark and West (2007). Our tests are one-sided following the literature on time series return predictability (e.g., Kelly and Pruitt 2013; Rapach, Strauss, and Zhou 2013; Huang, Jiang, Tu, and Zhou 2015).

We find that *persistent* flows deliver strong OOS return predictability. The OOS R^2 is large, particularly for hedge funds (6.59), but it is also sizable for mutual funds (3.65). In contrast, transient flows yield negative OOS R^2 for both fund types, indicating their inability to predict returns. Total hedge-fund flows exhibit no predictive power as their OOS R^2 is statistically insignificant and falls short of the 0.5 threshold for economic significance proposed by Campbell and Thomson (2008). When we do not orthogonalize flows with respect to factors,

¹⁴ Our choice of a 6-month moving window is motivated by the observation that many activities in asset management follow an annual cycle. See Section 3.1, 3.3 and Internet Appendix I4 for details. In any case, our finding that persistent flows possess predictive power for returns holds across various moving average windows (see Internet Appendix I4, Table I4-1).

¹⁵ Due to the well-known overfitting problem posed by OLS, we do not use flow variables jointly in a single multivariate OLS regression.

persistent flows' R^2 strengthens, while transient flows' remains negative (see Internet Appendix I8 Table I8-1). Thus, results based on residual flows provide a conservative representation of flows' OOS predictive power, and we will use residual flows in our subsequent OOS analyses.

Finally, we find that passive mutual-fund flows—including their persistent component—fail to predict factor returns OOS (see Internet Appendix I8 Table I8-4). This result highlights that only (persistent) active fund flows predict factor returns. Overall, these results reveal that persistent capital flows to active funds, especially to hedge funds, strongly predict factor returns OOS, whereas transient flows do not. As we will explain in Section 3.1, transient flows introduce noise into total flows, which reduces the predictive power of total flows.

2.2 Additional analyses

In Table 3, we perform five additional tests to assess the robustness of persistent (transient) flows' (lack of) predictive power. In Panels A and B, we present the results of various in-sample predictive regressions. Panel A displays in-sample coefficients estimated using predictive regressions spanning the full sample period. Panel B shows the average values of the in-sample coefficients and corresponding t -statistics obtained from the recursive regressions conducted monthly for the OOS tests of Table 2. Overall, the results of these in-sample predictive regressions closely mirror those of the OOS predictive regressions.

Table 3: Forecasting Regressions – Robustness and Horserace

The table examines the robustness of forecasting regressions. The control variables include lagged factor returns, Fama-French three factors, aggregate turnover, and aggregate Amihud illiquidity. We use 6-month moving averages (MA(6)) and first differences (Change) as low- and high-pass filters. The full sample covers 1994 to 2020. Based on Clark and West (2007), *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for R^2 statistics. Coefficients of interest that are statistically significant are bolded.

Panel A: Full In-Sample Predictive Regression Estimates

The panel reports the in-sample predictive regressions of factor returns on filtered fund flows for the entire sample period.

Fund Type	(1)	(2)	(3)	(4)
	Hedge Fund		Mutual Fund	
Flows	MA(6)	Change	MA(6)	Change
Beta	0.860 [2.70]	0.057 [0.27]	2.576 [2.97]	-0.042 [-0.05]
Controls	Yes	Yes	Yes	Yes
N	318	322	318	322
R^2	8.87%	6.26%	8.43%	6.22%

Panel B: Average Coefficient Estimates from Recursive In-Sample Predictive Regressions

The panel reports the average coefficient estimates of recursive in-sample predictive regressions used in OOS forecasts in Table 2.

Fund Type		(1)	(2)	(3)	(4)
		Hedge Fund		Mutual Fund	
Flows		MA(6)	Change	MA(6)	Change
Beta	Point Estimate	0.846	0.191	2.503	-0.036
	t-value (average)	[3.02]	[0.35]	[2.66]	[-0.27]

Panel C: OOS R^2 – Multi-Period Returns

This panel reports the R^2 of OOS regressions where the dependent variable is multi-period factor returns, $r_{t,t+j}$, where $j=0, 2$, and 5 . The forecasting horizon (h) is $j+1$.

Fund Type Forecasting Horizon (Months)	(1)	(2)	(3)	(4)	(5)	(6)
	Hedge Fund - MA(6)			Mutual Fund - MA(6)		
	1	3	6	1	3	6
$R^2(\%)$	6.59***	9.05***	8.42***	3.65***	6.36***	6.23***

Panel D: OOS R^2 – Self Predictability of Fund Flows

The panel reports the out-of-sample R^2 of filtered fund flows predicting their own future values.

Fund Type Flows	(1)	(2)	(3)	(4)
	Hedge Fund		Mutual Fund	
	MA(6)	Change	MA(6)	Change
$R^2(\%)$	5.36***	16.87***	11.42***	15.54***

Panel E: OOS R^2 – Horse Race

The panel reports out-of-sample R^2 of forecasting regressions of factor returns on various predictors as well as the encompassing test results using hedge-fund and mutual-fund moving averages. For each predictor, we estimate the following regression:

$$P_{t-1} = \gamma_0 + \gamma_1 r_{t-1} + \gamma_2' X_{t-1} + \varepsilon_{t-1},$$

where P_{t-1} denotes the predictor in month $t-1$, r_{t-1} is factor return in month $t-1$, and X_{t-1} is the set of control variables. Then, using the regression residuals, we estimate the following forecasting regression:

$$r_t = \alpha + \beta \varepsilon_{t-1} + \epsilon_t,$$

Finally, out-of-sample forecast for r_{t+1} is obtained from $\hat{\alpha} + \hat{\beta} \varepsilon_t$. The last two columns provide the results of encompassing tests. We provide the estimated weight (λ) on the predictive regression forecast based on HF-MA(6) (or MF-MA(6)) in a combination forecast based on each predictor. The statistical significance is based on the Harvey, Leybourne, and Newbold (1998) statistic for testing the null hypothesis that λ is zero. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Forecasting Vars		Book-to-Market	Characteristics	Lag Ret (MA(6))	Cum Mkt	Mkt Vol	Issue-Repurchase	Crowdedness
$R^2(\%)$	Point Estimate	0.66*	-1.08	-1.99	-0.28	0.29	1.17**	1.48*
Encompassing	λ (HF-MA(6))	1.00**	1.00***	1.00***	1.00***	1.00***	1.00**	0.90**
Tests	λ (MF-MA(6))	0.91***	1.00***	1.00***	0.79***	0.87***	0.93**	0.66**

In Panel C, we analyze the predictive power of flows for factor returns across horizons extending to six months (t to $t+5$). For persistent flows, the OOS R^2 increases with the horizon: in the case of hedge (mutual) funds, it grows from 6.59 (3.65) over one month to 8.42 (6.23) over six months. In contrast, transient flows do not predict returns over any horizon (untabulated). Thus, the predictive power of persistent flows stems from their ability to predict persistent movements in factor returns.¹⁶

In Panel D, we examine flows' self-predictability. To avoid a look-ahead bias, we use average flows estimated over the past six month to predict average flows over the next six month (consistent with the approach in Panel C for predicting future six-month cumulative returns). Here, the pattern is strikingly different: transient flows do predict future transient flows, and what's more, their ability to predict future transient flows is markedly

¹⁶ In Table I8-3 of the Internet Appendix, we re-estimated the OOS forecasts of Panel C, skipping one month. Thus, the dependent variables are the factor returns from month $t+1$ to $t+j$, where $j = 1, 3$, and 6 , while the independent variables are the 6-month backward moving average, at month $t-1$, of hedge and mutual fund flows. The results show that while moving average flows to hedge and mutual funds (at time $t-1$) have less predictive power for returns two months ahead (at $t+1$) compared to returns one month ahead (at t), both remain statistically significant at the 5% level. Additionally, skipping one month has a weaker impact on predictive power over longer horizons. Overall, forecasting returns with a one-month skip remains robust, thereby strengthening our OOS analysis.

stronger than persistent flows' ability to predict future persistent flows. Thus, transient flows inability to predict factor returns does not derive from those flows being themselves unpredictable.

Finally, in Panel E, we run a horse-race between persistent flows and established predictors from prior literature. We note that those predictors rely on economic mechanisms (reviewed in Internet Appendix I9) that differ from the underpinning we propose for persistent flows (which we model in Section 6). Specifically, we conduct a forecast encompassing test as in Harvey, Leybourne, and Newbold (1998). Its null hypothesis is that a given forecast contains all the relevant information present in a competing forecast (i.e., the given forecast encompasses the competitor's), while the alternative is that the competing forecast provides relevant information beyond what the given forecast contains. We start by forming an optimal forecast as a convex combination of the forecast built upon an existing predictor (i.e., a predictor introduced in a prior study) and of the forecast derived from persistent flows:

$$\hat{r}_{t:t+h} = (1 - \lambda)\hat{r}_{t:t+h}^{Existing} + \lambda\hat{r}_{t:t+h}^f.$$

Here $\hat{r}_{t:t+h}$ is the OOS optimal combination forecast, $\hat{r}_{t:t+h}^{Existing}$ represents the OOS predictive regression forecast based on the existing predictor, $\hat{r}_{t:t+h}^f$ corresponds to the OOS predictive regression forecast based on our persistent flow measure, and λ is a weight between 0 and 1. If $\lambda = 0$, then the optimal combination forecast excludes the forecast based on the persistent flow measure, implying that flows are redundant. If instead $\lambda > 0$, flows impart valuable information for forecasting factor returns, beyond the information already present in the existing predictor. Moreover, if λ is close to unity, then the optimal combination forecast excludes the predictive forecast based on the existing predictor, suggesting that the information contained in persistent flows subsumes completely that contained in the existing predictor. The significance of λ is assessed based on the null hypothesis that λ is equal to zero against the alternative that λ is greater than zero.

We compare our persistent flow measure to the following predictors documented in the literature: (1) *Book-to-Market*: the valuation ratio spread used in Haddad et al. (2020); (2) *Characteristics*: the spread of various anomaly characteristics considered in Kelly et al. (2021); (3) *Lag Ret* (MA(6)): the factor momentum concept of Ehsani and Linnainmaa (2022a); (4) *Cum. Mkt*: the cumulative market return considered in Cooper, Gutierrez, and Hameed (2004) and Daniel and Moskowitz (2016); (5) *Mkt Volatility*: the market return volatility considered in Daniel and Moskowitz (2016); (6) *Issue-Repurchase*: the issuer-repurchaser spread of Greenwood and Hansen

(2012); and (7) *Crowdedness*: the crowdedness measure of Lou and Polk (2021). We construct these predictors for anomalies comprising the SY set. Our focus on is the persistent component of hedge-fund flows, which directly relates to arbitrage capital. For completeness, we also report findings for mutual-fund flows.

Panel E shows that the estimates of λ corresponding to all the existing predictors we consider are large and statistically significant, indicating that none of the forecasts based on existing predictors renders redundant the forecast derived from persistent flows. In other words, persistent flows contain information incremental to these predictors for forecasting factor returns. Moreover, with the exception of crowdedness (the only predictor which, like ours, relates to arbitrageurs' trading activity), all λ estimates for persistent hedge-fund flows equal unity. These imply that the predictive regression forecast derived from persistent hedge-fund flows fully encompasses the forecasts based on these existing predictors. In the case of crowdedness, the λ estimate equals 0.9, which is also close to unity, indicating persistent flows contain considerable information that is not captured by crowdedness. Consistent with these findings, the top row of the table reveals that the highest OOS R^2 attained by existing predictors is 1.48 (for crowdedness), far lower than what persistent hedge-fund flows achieve (6.59). Finally, we also report in the bottom row the λ estimates for persistent mutual-fund flows. Those estimates are again statistically significant, but in most instances, smaller than for hedge funds. This implies that persistent mutual-fund flows contain unique information which need not completely subsume other predictors.

To summarize, these encompassing tests indicate that the persistent component of fund flows is a far stronger predictor of factor returns than are existing predictors. Indeed, it contains information incremental to these predictors in forecasting factor returns; and in but one instance, the information contained in persistent hedge-fund flows encompasses that provided by existing predictors. These results suggest that *persistent* arbitrage capital (within hedge funds) is highly informative about the *dynamics* of factor returns, specifically regarding the timing of factor profits (which differs from the unconditional magnitude of factor profits).

3. Frequency-based methodology for analyzing the sources of predictive power

To understand the *sources* of persistent flows' OOS predictability, we perform an in-sample analysis, as is commonly done in the literature. We adopt a frequency-domain approach which decomposes any stationary time series into an overlay of orthogonal components, each characterized by its persistence or frequency. The approach allows for a nonparametric, orthogonal decomposition of a series—which facilitates a *clean* analysis

and interpretation of the relation between flows and factor returns. In Internet Appendix I1, we provide more details and discuss how the notion of frequency differs from the related, but distinct, notions of holding period and turnover of anomaly strategies. In this section, we start by laying out a framework for analyzing the effect of hedge-fund flows on factor returns—the focus of our paper (Section 3.1), then extend this framework to account for the effect of mutual-fund flows (Section 3.2). We then discuss the close connection between the moving average/change filters used in Section 2 with the frequency-based filters (Section 3.3).

3.1 A framework for analyzing the relation of hedge-fund flows with returns

Through spectral analysis, any stationary time series can be expressed as a sum of orthogonal components. For hedge-fund flows, HF_t , we have (a similar decomposition holds for returns, r_t):

$$HF_t = HF_t^L + HF_t^H \quad \text{where } HF_t^L \perp HF_t^H; \quad (1)$$

here HF_t^L (HF_t^H) contains the components of HF_t that evolve within the low (high) frequency band, i.e., have frequencies of one cycle per year or less (more than one cycle per year). Using a one-year cutoff aligns with practice in the asset management industry as well as with the properties of moving average/change filters used in the OOS tests in Section 2 (see Internet Appendix I4). Notwithstanding, our main finding is robust to cutoffs of half a year, one year, one year and a half, and two years (See Internet Appendix I4 Table I4-2).

The low and high frequency components obtained for hedge-fund flows and returns through this decomposition have two useful properties. First, because they are orthogonal to one another, the contribution of frequencies in the band $K = \{L, H\}$ to the variance of HF_t is simply the ratio $\text{Var}(HF_t^K)/\text{Var}(HF_t)$. Components are also orthogonal across series, that is, $HF_t^K \perp r_t^{K'}$ for distinct frequency bands, K and K' . This implies that the contribution of frequencies in band K to the covariance between HF_t and r_t is given by the ratio $\text{Cov}(HF_t^K, r_t^K)/\text{Cov}(HF_t, r_t)$. A further consequence is that the coefficient from regressing r_t on HF_t equals the variance-weighted sum of the coefficients in each band K , $\frac{\text{Var}(HF_t^K)}{\text{Var}(HF_t)}\beta_L + \frac{\text{Var}(HF_t^H)}{\text{Var}(HF_t)}\beta_H$, where these coefficients are estimated from a *contemporaneous* regression of r_t^K on HF_t^K :

$$r_t^K = \alpha_K + \beta_K HF_t^K + \varepsilon_t^K; \quad (2)$$

here ε_t^K is uncorrelated with HF_t^K and i.i.d.. Importantly, covariances and regression coefficients may be negative at some frequencies even though they are positive for the total series.

As for the second useful property of the spectral decomposition, it turns out that the components of hedge-fund flows and returns can be described by an AR1 process. For HF_t , it reads (and similarly for r_t):

$$HF_t^K = \rho_{HF}^K HF_{t-1}^K + v_t^K, \quad (3)$$

where v_t^K is white noise. This regression's R^2 , $(R^2)^{HF_{t-1}^K \rightarrow HF_t^K} = (\rho_{HF}^K)^2$, measures flows' self-predictability over the frequency band K . This characterization allows to link the predictive regressions of returns on flows to their contemporaneous regressions. Indeed, substituting the AR1 expression for flows (3) into the contemporaneous regression (2) leads to the (band- K) predictive regression:

$$r_t^K = \alpha_K + \beta_K \rho_{HF}^K HF_{t-1}^K + \varepsilon'_t. \quad (4)$$

The regression's (predictive) R^2 can be expressed as:

$$(R^2)^{HF_{t-1}^K \rightarrow r_t^K} \equiv \frac{(\beta_K \rho_{HF}^K)^2 \text{Var}(HF_t^K)}{\text{Var}(r_t^K)} = \frac{(\beta_K)^2 \text{Var}(HF_t^K)}{\text{Var}(r_t^K)} (\rho_{HF}^K)^2 = (R^2)^{HF_t^K \rightarrow r_t^K} (R^2)^{HF_{t-1}^K \rightarrow HF_t^K} \quad (5)$$

where the first R^2 on the right-hand side of (5) measures the contemporaneous explanatory power of hedge-fund flows for returns over frequency band K (from regression (2)), and the second R^2 measures flows' self-predictability over that frequency band.

Because of orthogonality, the R^2 of the regression of *total* returns, r_t , on lagged hedge-fund flows, HF_{t-1}^K (or on contemporaneous flows, HF_t^K) is equal to the R^2 of the (band- K) regression of r_t^K on HF_{t-1}^K (or on HF_t^K), scaled by the proportion of returns' variance generated over the band K :

$$(R^2)^{HF_t^K \rightarrow r_t} = \frac{\text{Var}(r_t^K)}{\text{Var}(r_t)} (R^2)^{HF_t^K \rightarrow r_t^K} \text{ for } \tau = \{t, t-1\}. \quad (6)$$

Intuitively, if flows explain or predict returns over a frequency band, then they also explain or predict total returns, provided that return variations over that band represent a sizable proportion of returns' total variations. Moreover, if one component of flows has strong predictive power for returns while the other weak power, then the predictive power of *total* flows for total returns is weaker than that of the former component because of the noise the latter component creates. As an illustration (matching what we find in the data), suppose that high-frequency flows have weak predictive power (i.e., $\text{Cov}(r_t, HF_{t-1}^H) \approx 0$) and large variability (i.e., $\text{Var}(HF_{t-1}^H)$ is big). Then, the coefficient b , in the predictive regression of total returns on total hedge-fund flows, is small:

$$b = \frac{Cov(r_t, HF_{t-1})}{Var(HF_{t-1})} = \frac{Cov(r_t, HF_{t-1}^H + HF_{t-1}^L)}{Var(HF_{t-1}^H + HF_{t-1}^L)} = \frac{Cov(r_t, HF_{t-1}^H) + Cov(r_t, HF_{t-1}^L)}{Var(HF_{t-1}^H) + Var(HF_{t-1}^L)} \approx \frac{Cov(r_t, HF_{t-1}^L)}{Var(HF_{t-1}^H) + Var(HF_{t-1}^L)} \ll \frac{Cov(r_t, HF_{t-1}^L)}{Var(HF_{t-1}^L)}. \quad (7)$$

Similarly, the contemporaneous explanatory power of flows for returns over the frequency band K , $(R^2)^{HF_t^K \rightarrow r_t^K}$, contains information about the contributions of HF_t^K to the autocorrelation and variance of r_t^K . Again, combining the AR1 expression for flows (3) with the contemporaneous regression (2) implies that the first-order autocorrelation and variance of r_t^K are given, respectively, by:

$$\rho_r^K = \rho_{HF}^K (R^2)^{HF_t^K \rightarrow r_t^K} \quad \text{and} \quad Var(r_t^K) = \beta_K^2 Var(HF_t^K) + Var(\varepsilon_t^K), \quad (8)$$

where ρ_r^K and ρ_{HF}^K denote the serial correlations of r_t^K and HF_t^K , respectively. So, HF_t^K 's relative contributions to r_t^K 's serial correlation and variance can be measured, respectively, as:

$$\frac{\rho_{HF}^K}{\rho_r^K} (R^2)^{HF_t^K \rightarrow r_t^K} \quad \text{and} \quad (R^2)^{HF_t^K \rightarrow r_t^K}. \quad (9)$$

Thus, the larger is $(R^2)^{HF_t^K \rightarrow r_t^K}$, the closer are the serial correlation and variance structures of HF_t^K to those of r_t^K . Hence, HF_t^K 's contributions to the serial correlation and variance of total returns, r_t , equal, respectively:

$$\frac{Var(r_t^K) \rho_{HF}^K}{Var(r_t) \rho_r} (R^2)^{HF_t^K \rightarrow r_t^K} \quad \text{and} \quad \frac{Var(r_t^K)}{Var(r_t)} (R^2)^{HF_t^K \rightarrow r_t^K} \quad (10)$$

Hence, the serial correlation and variance structures of HF_t^K determine those of total returns, i.e., determine factor autocorrelation and factor volatility, provided that $(R^2)^{HF_t^K \rightarrow r_t^K}$ is high and the return variance over the frequency band K accounts for a sizable proportion of the variance of total return r_t .

The relations derived in this section extend naturally to a multivariate setting that includes control variables (see Internet Appendix I2), where we redefine the variance ratio, $\frac{(\beta_K)^2 Var(HF_t^K)}{Var(r_t^K)}$, as partial R^2 .

3.2 Incorporating mutual-fund flows

We introduce mutual funds into the framework of Section 3.1. Equation (2), which describes the *contemporaneous* flow-return relationship over the frequency band K , is extended to account for the *joint* effect of hedge- and mutual-fund flows on factor returns as follows:

$$r_t^K = \alpha_K + \beta_K HF_t^K + \lambda_K MF_t^K + \varepsilon_t^K, \quad (11)$$

where HF_t^K and MF_t^K denote the band- K component of hedge- and mutual-fund flows, and ε_t^K is uncorrelated

with both types of flows and i.i.d.. Next, we show how this equation provides insights into the predictive power of hedge-fund flows for returns. In the *predictive* regression,

$$r_t^K = \alpha_K + b_K^{HF} HF_{t-1}^K + \varepsilon_t^{HF,K}, \quad (12)$$

the coefficient b_K^{HF} can be expressed as:

$$\begin{aligned} b_K^{HF} &= \frac{Cov(\beta_K HF_t^K + \lambda_K MF_t^K + \varepsilon_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)} = \beta_K \frac{Cov(HF_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)} + \lambda_K \frac{Cov(MF_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)} \\ &= \beta_K \rho_{HF}^K + \lambda_K \frac{Cov(MF_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)}.^{17} \end{aligned} \quad (13)$$

To be concise, we focus on the low-frequency band (i.e., $K = L$) for the remainder of this section. Indeed, as we reported in Section 2—and shall confirm through spectral analysis in Section 4 and theoretically in Section 6, transient flows have no predictive power.

The first term of Equation (13), $\beta_K \rho_{HF}^K$, represents hedge funds' *own-flow effect*. As we document later, the parameter β_K is positive at low frequency ($\beta_L > 0$). This implies that positive flows to hedge funds are accompanied by positive factor returns—corresponding to a correction of mispricing (see Section 1.1). Hence, consistent with hedge funds being smart money, β_K measures how much mispricing is corrected *contemporaneously* by one unit of hedge-fund flows. The parameter ρ_{HF}^K captures the self-predictability of hedge-fund flows: one unit of flows is followed by an additional ρ_{HF}^K units in the subsequent period ($\rho_{HF}^L > 0$). Thus, $\beta_K \rho_{HF}^K$ measures how much mispricing hedge-fund flows correct in the subsequent period ($\beta_L \rho_{HF}^L > 0$).

The second term of Equation (13), $\lambda_K \frac{Cov(MF_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)}$, represents a *cross-flow effect*, that is, the effect on mispricing of mutual-fund flows that are expected to follow hedge-fund flows. As we report later, the parameter λ_K , which represents the contemporaneous effect of mutual fund-flows on factor returns, is negative at low frequency ($\lambda_L < 0$). This indicates that positive flows to mutual funds are associated with negative factor returns, i.e., with an exacerbation of mispricing, consistent with mutual funds being dumb money. In the data, hedge-fund flows have a positive covariance with future mutual-fund flows, implying that the cross-flow effect counters the own-flow effect. However, the covariance is small and the variance large, so $\frac{Cov(MF_t^K, HF_{t-1}^K)}{Var(HF_{t-1}^K)} \approx 0$ and the

¹⁷ $Cov(\varepsilon_t^K, HF_{t-1}^K) = 0$ as ε_t^K represents shocks to returns unrelated to flows. This is confirmed by the estimation of Equation (11), where the resulting residual shows no significant correlation with lagged flows.

cross-flow effect is weak. Therefore, the predictive power of hedge-fund flows derives primarily from their self-predictability and we expect $b_L^{HF} > 0$ in Equation (13). Also, Equation (5) (which ignores any cross-flow effect) captures the bulk of the predictive power of hedge-fund flows for returns.

The case of mutual funds (analyzed in Internet Appendix I3) is more complex because their cross-flow effect cannot be ignored: mutual-fund flows are followed by more mutual-fund flows that exacerbate mispricing, but also by hedge-funds flows of comparable magnitude that correct it. As a result, the coefficient in predictive regressions can be positive or negative, and the predictive power is likely weaker than hedge funds'. In any case, our focus is on arbitrage capital, and we use mutual funds merely as a comparison.

3.3 The connection between moving average/change and frequency-based filters

The moving/change filters employed for the predictive regressions in Section 2 closely resemble the frequency-based filters with a one-year cycle as cutoff. Indeed, the persistent (transient) components extracted from those filters are highly correlated (see Section 4.1 and Table 1, Panel D), and over 90% of the variations in the moving average (change) series belong to the low- (high-)frequency band (see Internet Appendix I4, Table I4-3). For a comprehensive discussion, please refer to Internet Appendix I4.

Yet, each approach comes with its own pros and cons. Moving average/change filters provide a simple and intuitive way to coarsely generate persistent and transient components to use in contemporaneous regressions as well as OOS and in-sample predictive regressions. But those components are correlated, and moreover, sensitive to parametric assumptions, which makes attribution analysis difficult. On the other hand, spectral analysis offers an orthogonal and nonparametric decomposition, but is ill-suited for predictive regressions due to its inherent look-ahead bias. For these reasons, we combine both approaches, employing the first set of filters for OOS predictive regressions (Section 2) and the second for contemporaneous regressions and attribution (Section 4).

To further connect the two sets, we present, in Internet Appendix I5, results obtained through a third approach—a Vector Autoregressive (VAR) model. This model extracts persistent and transient flow components in a sophisticated and structured, though not orthogonal, fashion. The model allows for both predictive and contemporaneous regressions, and yields results similar to those in Sections 2 and 4. Therefore, these findings confirm the robustness of our key conclusions which we obtained through alternative approaches.

4. The contemporaneous relation between flows and factor returns

Table 3, Panel D indicates that persistent flows are less *self*-predictable than transient flows. From Equation (5), this implies that the main source of their predictive power for factor returns must be their (contemporaneous) explanatory power for factor returns. In the remainder of the paper, we shed light on this explanatory power (corresponding to Equation (2)) and on its causes.

4.1 Spectral decomposition of factor returns and flows

We first describe the spectral decompositions of factor returns and flows. Before proceeding, we perform unit-root tests to confirm that these series are stationary—a prerequisite for a Fourier transformation (Table I1-1 of the Internet Appendix I1). Variables with the “-LOW” (“-HIGH”) suffix represent time series reconstructed from frequencies lower than or equal to (higher than) one cycle per year. Because flow variables are measured monthly, our highest frequency is six cycles per year.

Table 1 presents summary statistics. In Panel A, the average monthly return on the SYI is significantly positive at the 1% level with large magnitudes, indicating that the strategy is profitable. Panel B indicates that factor returns and flows are positively autocorrelated. Decomposing the series uncovers contrasting patterns: for both series, autocorrelations are positive at low frequency—as expected—but negative at high frequency.

Panel C reports the ratio of the variance of the decomposed series to that of the total series—which, given the orthogonality property discussed in Section 3.1, represents the contributions of each component to the total variance. The panel reveals that the series differ markedly in their frequency structures. Factor returns vary about three times more at high frequency than they do at low frequency. In contrast, hedge-fund flows are driven almost equally by low- and high-frequency fluctuations, and mutual- (passive-) fund flows are mostly driven by low- (high-)frequency fluctuations.¹⁸ Thus, comparatively, mutual fund investors are the “slowest” in providing capital in that their flows are most persistent, followed by hedge and passive fund investors. This ranking is broadly in line with the characteristics of the clientele that these funds serve. For instance, mutual funds have many slow-moving pension fund investors whereas passive funds may have many fast-moving liquidity traders.

Panels B and C also indicate that the moving average and change series used in Section 2 display broadly similar properties to the frequency-based series. Additionally, Panel D shows that the correlations between the persistent (transient) components ranges from 0.89 to 0.98 (0.69 to 0.74).

¹⁸ Passive mutual fund flows are mostly driven by high-frequency fluctuations (54.1% of total variance).

4.2 Explanatory power of flows for factor returns: The low-frequency tilt

Table 4 presents estimates from contemporaneous regressions of low- (high-)frequency factor returns on low- (high-)frequency flows, controlling for the same five risk and liquidity factors employed for constructing residual flows in OOS tests. The regressions and controls follow those in Akbas et al. 2015. We focus here on partial R^2 s (columns labeled “partial R^2 ”) which measure a regressor’s explanatory power after removing the effect of other regressors. At low frequency, hedge- (mutual-)fund flows explain 9.6% (16.9%) of the total variation in SY Y returns—far more than other factors, the market included—representing 21% (37%) of regressors’ total contributions. At high frequency, their explanatory power drops to 0.0% (2.9%). Compared to recent prominent models—those of Daniel, Hirshleifer, and Sun (2020), Fama and French (2015), and Hou, Xue, and Zhang (2015), flows have far more explanatory power at low frequency than any of the factors in those models (Table I8-5 of the Internet Appendix I8). Those findings reveal a powerful *tilt toward low frequencies* in the relation between fund flows and factor returns.

Table 4: Factor–Flow Relation

This table reports the results of contemporaneous time-series regressions of long-short factor returns on fund flows. The dependent variables are the low- and high-frequency components of long-minus-short returns in month t of a composite factor, SY Y, the return of the long-minus-short strategy based on anomalies in SY Y (2012). The main independent variables are the low- and high-frequency components of MF and HF in month t , where MF and HF are the percentage flows to active mutual funds and hedge funds. For each model, the first column shows the coefficient estimate of each variable and the corresponding t -value. The second column reports the contribution of each variable to the variance of factor returns, calculated from partial R^2 , and the ratio of the contribution to the sum of all contributions. t -statistics are calculated based on Newey-West standard errors with 13 lags. The sample period is 1994–2020.

	(1)		(2)	
	FREQ = LOW		FREQ = HIGH	
	Beta	Partial R^2 / (% Contribution)	Beta	Partial R^2 / (% Contribution)
HF-FREQ	0.911 [4.14]	9.6% (21%)	0.085 [0.41]	0.0% (0%)
MF-FREQ	-4.029 [-4.32]	16.9% (37%)	-2.527 [-2.80]	2.9% (10%)
MKT	-0.138 [-2.97]	5.4% (12%)	-0.361 [-4.80]	10.3% (36%)
Amihud	0.374 [2.35]	5.1% (11%)	-0.157 [-0.82]	0.3% (1%)
Turnover	-0.014 [-1.77]	5.9% (13%)	-0.001 [-0.19]	0.0% (0%)
HML	0.116 [1.49]	2.1% (5%)	0.249 [1.68]	2.8% (10%)
SMB	-0.014 [-0.33]	0.0% (0%)	-0.513 [-6.87]	11.5% (40%)
N	324		324	
R^2	31.1%		34.5%	

Notably, that variables have more explanatory power at lower frequency is not mechanical. For instance, some factors, such as the market and size factors, have more explanatory power at high frequency than at low

frequency. Moreover, the series are far from random walks, so their low-frequency connection is not spuriously caused by a very low frequency component such as a trend.

Table 5: The Low Frequency Tilt

The table quantifies the low frequency tilt in the flow effect on the SYF factor return. Panel A reports the contribution of flows to variations in factor returns. Columns (4) and (6) report the explanatory power (R^2) of flows f_t^K for returns r_t^K over the frequency band K ($=L, H$), and for raw return r_t , respectively. Column (7) reports the economic magnitudes of the flow-return regression in Table 2, calculated as $\beta_K \times \sigma_r^K / \sigma_r^K$, where σ_r^K and σ_r^K denote, respectively, the standard deviations of flows and the factor return over the band K , and β_K is the coefficient estimate from regressing the factor return on flows, over the band K , as reported in Table 2. Panel B reports the contribution of flows to the serial correlation of the factor return, calculated as $\rho_r^K \times (\beta_K)^2 \times \text{Var}(f_t^K) / \text{Var}(r_t^K)$ divided by ρ_r^K . The “L/H” row reports the cross-frequency ratio for the corresponding column, defined as the ratio of the parameter in that column at low (L) frequency to its magnitude at high (H) frequency. The sample period is 1994–2020.

Panel A: Contribution of Flows to the Variations in Factor Returns

Fund Type	Frequency	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Manager	Investor		$(R^2)(f_t^K \rightarrow r_t^K)$	Importance of r_t^K in r_t	$(R^2)(f_t^K \rightarrow r_t)$	Econ. Magnitude	Manager vs. Investor
		β_K	σ_r^K	σ_r^K	$(\beta_K)^2 \text{Var}(f_t^K) / \text{Var}(r_t^K)$	$\text{Var}(r_t^K) / \text{Var}(r_t)$	(4)×(5)	$\beta_K \sigma_r^K / \sigma_r^K$	(1)/(2)
HF	K=L	0.911	1.14%	2.85%	13.4%	22.1%	3.0%	36.6%	
	K=H	0.085	1.10%	5.35%	0.0%	77.9%	0.0%	1.7%	
	L / H	10.72	1.04		442.19		125.24	21.03	10.27
MF	K=L	-4.029	0.45%	2.85%	40.9%	22.1%	9.0%	-64.0%	
	K=H	-2.527	0.32%	5.35%	2.3%	77.9%	1.8%	-15.0%	
	L / H	1.59	1.42		18.07		5.12	4.25	1.12
Ratio HF/MF		6.72	0.74		24.47		24.47	4.95	9.14

Panel B: Contribution of Flows to the Serial Correlation of Factor Returns

Fund Type	Frequency	(1)	(2)	(3)	(4)	(5)	(6)
					ρ_r^K due to f_t^K		Contribution of f_t^K to ρ_r^K
		ρ_r^K	β_K	σ_r^K	$\rho_r^K \times (\beta_K)^2 \times \text{Var}(f_t^K) / \text{Var}(r_t^K)$	ρ_r^K	(4)/(5)
HF	K=L	0.967	0.911	1.14%	0.130	0.955	13.6%
	K=H	-0.126	0.085	1.10%	0.000	-0.103	0.0%
	L / H		10.72	1.04	-3386.47		366.92
MF	K=L	0.979	-4.029	0.45%	0.401	0.955	42.0%
	K=H	-0.054	-2.527	0.32%	-0.001	-0.103	1.2%
	L / H		1.59	1.42	-326.46		35.37
Ratio HF/MF			6.72	0.74	10.37		10.37

To quantify the low-frequency tilt and compare it across fund types, we report in Panel A of Table 5 (row labeled “L/H”, Columns 4 and 6) the ratios of flows’ contribution to the total variation of the SYF factor across frequencies. The ratios in Column 4, $(R^2)^{f_t^L \rightarrow r_t^L} / (R^2)^{f_t^H \rightarrow r_t^H}$, imply that the explanatory power of hedge- (mutual-) fund flows for factor returns is significantly larger at low frequency than at high frequency (e.g., for hedge funds, low vs. high is 13.4% vs. 0.0%). The ratios in Column (6), $(R^2)^{f_t^L \rightarrow r_t} / (R^2)^{f_t^H \rightarrow r_t}$, which pertain to *total* returns, also reveal a strong low-frequency tilt, especially for hedge funds. The weakening is caused by the fact that low-frequency variations account for a smaller, but still sizeable, fraction of total return variations (22.1%) than do high-frequency variations (Column (5)). Comparing fund types, hedge funds stand out: their

low-frequency tilt is multiple times stronger than for mutual funds (Row “HF/MF” in Column (4) or (6)).

4.3 Sign and significance of the flow-return relation

The sign of the coefficient estimates in the flow-return regressions offers insights on funds’ strategies even if it does not affect R^2 . Table 4 shows that it is positive across frequencies for hedge funds but negative for mutual funds. The coefficient estimates are statistically significant throughout, except for hedge funds at high frequency. Moreover, their magnitude tends to weaken at high frequency, in line with the low-frequency tilt.

Correcting (exacerbating) mispricing in an overvalued stock entails taking a short (long) position and so, by construction, is associated with a positive (negative) return on the hedge portfolio (Kokkonen and Suominen 2015, Akbas et al. 2015). Hence, our findings indicate that net flows to hedge funds contribute to mispricing corrections—thus constituting smart money—but *only significantly so at low frequency*. In contrast, net flows to mutual funds contribute to mispricing—and appear to be dumb money—over the entire spectrum. The signs are consistent with the intuition derived in Section 3.2. In Sections 5-7, we will present causal evidence, a theory, and direct trading evidence in support of these interpretations.

We return to Panel A of Table 5 to evaluate the economic magnitude of the effect of flows on factor returns. Column (7) quantifies the effect on returns (expressed in units of their standard deviation) of a 1-SD increase in flows, $\beta_K \times \sigma_f^K / \sigma_r^K$. At low frequency, a 1-SD increase in hedge- (mutual-) fund flows corrects (exacerbates) mispricing by 36.6% (64.0%) of a SD. At high frequency, in contrast, it corrects (exacerbates) mispricing by merely 1.7% (15.0%) of a SD. Row “L/H” reports the ratio of those magnitudes across frequencies, $(\beta_L \times \sigma_f^L / \sigma_r^L) / (\beta_H \times \sigma_f^H / \sigma_r^H)$. The ratios imply that hedge and mutual funds, respectively, correct and exacerbate mispricing at low frequency, respectively, 21.03 and 4.25 times more than they do at high frequency.

Comparing fund types, the ratio is 4.95 times larger for hedge funds than for mutual funds. This implies that, at low frequency, a 1-SD increase in hedge-fund flows can correct 57% ($=36.6\%/64.0\%$) of the mispricing entailed by a 1-SD increase in mutual-fund flows, vs. only 11% ($=1.7\%/15.0\%$) at high frequency. This finding again points to the low-frequency tilt being markedly stronger for hedge funds than for mutual funds.

Importantly, mutual funds manage 30 times more assets than hedge funds over our sample period on average. This stark difference underscores the effectiveness of hedge funds in shaping mispricing per dollar invested, further justifying our focus on them.

4.4 Drivers of the low-frequency tilt

Difference in $(R^2)^{f_t^K \rightarrow r_t}$ across frequencies must be driven by differences in the regression coefficient and/or in the volatility of flows (the regressor). The first driver, the coefficient, is larger at low frequency than at high frequency by a factor 10.7 (1.6) for hedge (mutual) funds (see the ratio β_L/β_H in Column (1) of Table 5, Panel A). The second driver, volatility, is larger by a factor 1.04 (1.42) (see the ratio σ_f^L/σ_f^H in Column (2)). Thus, the low-frequency tilt stems entirely from differences in the regression coefficient at hedge funds, and from differences in both the regression coefficient and the volatility of flows at mutual funds

The regression coefficient and the volatility of flows can be interpreted as reflecting the behavior of fund managers (who deploy capital into stocks) and investors (who supply capital), respectively. A ratio β_L/β_H larger (smaller) than one means that flows have a larger (smaller) effect on returns at low frequency than at high frequency. Accordingly, the estimates in Column (1) suggest that managers, especially at hedge funds, actively attenuate (amplify) the effect of high-(low-)frequency flows; in other words, they aggressively invest persistent capital flows in mispriced stocks while repurposing transient flows. A larger ratio σ_f^L/σ_f^H can be interpreted as investors supplying capital more “slowly”. The estimates in Column (2) suggest that investors supply capital to mutual funds “slowly” since low-frequency flows explain a third more of the total variations in flows than do high-frequency flows. This is not the case for hedge funds for which variations in flows are equally driven by low- and high frequency components.

Comparing the cross-frequency ratios across fund types indicates that hedge fund managers attenuate the effect of high-frequency flows about 6 times ($= 10.72/1.59-1$) more than do mutual fund managers, and that mutual fund investors supply capital one-third more slowly than do hedge fund investors ($= 1.42/1.04-1$). Thus, managers are responsible for the low-frequency tilt almost entirely at hedge funds, and marginally more than investors at mutual funds.

Finally, we examine passive funds. Since passive managers have no discretion in dealing with flows, they serve as a useful benchmark for understanding the effect of active management on the flow-return relation. Using the same regressions as in Table 4, we find that passive fund flows are insignificantly related to factor returns across frequencies, and display no low-frequency tilt (see Internet Appendix I8, Table I8-6). This supports our interpretation that active management is the source of the low-frequency tilt. Since passive funds typically track

broad indexes rather than take long-short market-neutral positions in anomaly portfolios, we sharpen our test by estimating the effect of their flows, not on anomaly-based factor returns, but on two market return proxies: the CRSP value-weighted market returns and the return on the S&P 500 index.

Table 6: Flows to Passive Funds

The table reports the effects of flows to passive mutual funds on stock market returns. Passive funds are identified by names, following Appel, Gormley, and Keim (2016). Panel A examines the flow-return relation using, as dependent variables, the low- and high-frequency components of the value-weighted market return (MKT) and returns to S&P500 index (S&P). Panel B shows the contribution of flows to variations in the returns of MKT and S&P. The economic magnitude of the flow effect over frequency band $K=(L, H)$, reported in Column (7), is calculated as $\beta_K \times \sigma_f^K / \sigma_r^K$, where σ_f^K and σ_r^K denote, respectively, the standard deviations of flows and market returns over the band K , and β_K is the coefficient estimate from regressing market returns on flows over the band K , as reported in Panel A. The “L/H” row reports the cross-frequency ratio for the corresponding column, defined as the ratio of the parameter in that column at low (L) frequency to its magnitude at high (H) frequency. The control variables include the aggregate Amihud measure, the aggregate turnover, HML and SMB. t -statistics are calculated based on Newey-West standard errors with 13 lags. The sample period is 1994–2020.

Panel A: Flow Return Regressions

	(1)		(2)		(3)		(4)	
	FREQ = LOW				FREQ = HIGH			
Variables	MKT-LOW		S&P-LOW		MKT-HIGH		S&P-HIGH	
Passive-FREQ	-0.119	[-0.22]	-0.062	[-0.12]	1.130	[3.80]	1.129	[3.87]
Controls	Yes		Yes					
N	324		324		324		324	
R ²	6.8%		7.4%		11.3%		7.0%	

Panel B: Economic Significance

Returns	Frequency	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Manager	Investor		(R ²)(f _t ^K → r _t ^K)	Importance of r _t ^K in r _t	(R ²)(f _t ^K → r _t)	Econ. Magnitude	Manager vs. Investor
		β_K	σ_f^K	σ_r^K	$(\beta_K)^2 \text{Var}(f_t^K) / \text{Var}(r_t^K)$	$\text{Var}(r_t^K) / \text{Var}(r_t)$	(4)×(5)	$\beta_K \sigma_f^K / \sigma_r^K$	(1)/(2)
MKT	K=L	-0.119	0.75%	1.87%	0.2%	17.5%	0.0%	-4.8%	
	K=H	1.130	0.79%	4.06%	4.9%	82.5%	4.0%	22.1%	
	L / H	-0.10	0.95		0.05		0.01	-0.22	-0.11
S&P	K=L	-0.062	0.75%	1.81%	0.1%	17.7%	0.0%	-2.6%	
	K=H	1.129	0.79%	3.91%	5.2%	82.3%	4.3%	22.9%	
	L / H	-0.06	0.95		0.01		0.00	-0.11	-0.06

Panel A of Table 6 reveals that passive funds have a strong effect at high frequency and none at low frequency, so display a *high*-frequency tilt. In fact, for those funds, the ratio of economic magnitudes across frequencies, $(\beta_L \times \sigma_f^L / \sigma_r^L) / (\beta_H \times \sigma_f^H / \sigma_r^H)$ is close to zero (-0.22 and -0.11 in rows labeled “L/H” and Column (7) of Panel B). Breaking up the ratio indicates that their low value does not stem from differences in volatility (the volatility ratio in Column (2) equals 0.95) but from differences in the regression coefficient (the beta ratios equal -0.10 and -0.06 in Column (1))—in other words not from investors but from managers. With passive managers required to invest flows in benchmark stocks without discretion or delay, these low ratios should reflect the bigger price impact of higher frequency flows: inflows (outflows) lead to purchases (sales) that push up (down) stock prices more if they occur at higher frequency, because market makers require a larger compensation

for accommodating orders over a shorter period (e.g., Kyle et al. 2017). These findings highlight the central role played by active management for the low-frequency tilt; without it, there would (probably) be no such effect.

The contrast with passive funds makes the low-frequency tilt displayed by active funds all the more striking, given that it occurs despite the provision of liquidity being limited. Such limits most likely result in a higher price impact for higher-frequency flows and in a high-frequency tilt.¹⁹

Overall, guided by the connection between contemporaneous and predictive regressions theorized in Section 3.1 and 3.2, Section 4 sheds light on why persistent, but not transient, flows predict factor returns. Our results suggest that managers, especially those at hedge funds, play a central role in attenuating the effect of high-frequency flows on factor returns. In the next three sections, we support this interpretation by (i) addressing concerns about the endogeneity of flows, (ii) modeling the mechanism at work, and (iii) testing ancillary predictions implied by the model.

5. A model of the predictive power of (persistent) flows for returns

We present here a limits-to-arbitrage model describing how the ability of *sector-wide* fund flows to predict factor (i.e., mispriced assets') returns depends on the persistence of flows.

5.1 Setup

The model features four periods, $t = 0, 1, 2, 3$, two assets, and two types of agents.

Assets. A riskless asset (cash) in perfectly elastic supply has a return normalized to zero. A risky asset, representing a factor, has a unit supply, a price p_t and a return $r_t = (p_t - p_{t-1})/p_{t-1}$ in period t . At date $t = 3$, the asset's price converges to its fundamental value, $V = v + \varepsilon_1 + \varepsilon_2 + \varepsilon_3$, with certainty, where v is known at $t = 0$ and ε_τ ($\tau = 1, 2, 3$) is a random shock observed at time τ . Thus, the expectation of the fundamental value evolves according to $E_t(V) = v + \sum_{\tau=1}^t \varepsilon_\tau$ for $t \geq 1$. The shocks ε_τ are identically and independently distributed over time with mean zero and take values in the segment $[\underline{\varepsilon}, \bar{\varepsilon}]$ where $\underline{\varepsilon} < 0$ and $\bar{\varepsilon} > 0$.

Agents. Two types of agents participate in the market. *Noise traders* generate or exacerbate anomalies. They demand $(E_t(V) - s_t)/p_t$ units of the asset in period t where s_t is a demand shock. We assume that s_t is positive—it reflects pessimism—and induces the stock to be underpriced.²⁰ For ease of exposition, we interpret

¹⁹ In addition to our passive funds evidence, studies in microstructure, both theoretical and empirical, demonstrate that faster trading induces a bigger price impact (e.g., Kyle et al. 2017). Hence, to the extent that higher-frequency flows entail adjusting positions faster, they are likely to generate a bigger price impact and, hence, lower cross-frequency ratios.

²⁰ The results are similar for a negative demand shock causing the asset to be overpriced, inducing arbitrageurs to short the asset.

the demand shock as mispricing. But it could be viewed as a compensation for risk that noise traders require for holding the asset. There is no uncertainty with respect to the demand shock: it is assumed to persist until period 2 ($s_0 = s_1 = s_2 \equiv s > 0$), and then disappears in period 3 ($s_3 = 0$). The following assumption ensures that prices in periods 1 and 2 are non-negative.

Assumption 1. $s < v + 2\varepsilon$.

The other agents are risk-neutral *arbitrageurs* (e.g., hedge funds). A mass one of identical, such arbitrageurs indexed by i enter the market in period 1 and compete. Arbitrageur i has $A_{t,i}$ assets under management (AUM). She invests those AUM in the risky asset or in cash, with the objective of maximizing expected next period AUM—so she behaves myopically.²¹ We denote by $\theta_{t,i}$ the fraction of AUM invested in the risky asset by arbitrageur i in period t . To account for the fact that arbitrage capital is limited, we assume that *arbitrageurs cannot borrow*, which restricts $\theta_{t,i}$ to be below 1. Given the zero riskfree rate, the fund's return between dates $t - 1$ and t is equal to $\theta_{t-1,i} r_t$.

Flows. We assume that arbitrageurs are subject to (exogenous) flows of capital.²² How flows are distributed across funds does not matter in the model, so we assume that they are identical across funds, and coincide with aggregate flows into the arbitrage sector. We denote f_t the net flow to fund i between periods $t - 1$ and t , as a fraction of its period- $t - 1$ AUM. At the time they enter the asset market (date 1), f_1 is known to arbitrageurs, but f_2 is random: with probability q , flows persist and $f_2 = f_1$; with probability $1 - q$, they reverse and $f_2 = -f_1$. Flows f_2 are assumed independent from shocks ε_t . The probability q is the key parameter of the model; the higher q , the more persistent flows are. In particular, *flows are persistent (transient)* if $q > 1/2$ ($q < 1/2$). Flows in period 3 play no role in the model so are set to zero.

AUM in period t depend on each fund's return and net flows between $t - 1$ and t , and are given by $A_{t-1,i} (1 + \theta_{t-1,i} r_t + f_t)$ where $A_{0,i}$ and $\theta_{0,i}$ are normalized to 1 and 0, respectively (so $A_{1,i} = 1 + f_1$).

²¹ We confirm that the model's predictions are largely insensitive to the assumption on managers' horizon. In Internet Appendix I11, we present a version of the model in which managers maximize a long-term objective, namely expected period-3 AUM. This model leads to predictions that are similar to those of the model with myopic arbitrageurs. In addition, it yields the following expression for the SDF: $m_{2,i} \equiv \frac{\varphi_{2,i}}{E_1(\varphi_{2,i}) + \varphi_{1,i}/A_{1,i}}$ where $\varphi_{t,i}$ denotes Lagrange multiplier on arbitrageur i 's borrowing constraint, $\theta_{t,i} \leq 1$, and represents the marginal value of her wealth (AUM) in period t . The asset price is then determined by the traditional equation $E_1(m_{2,i} R_2) = 1$ where $R_2 \equiv 1 + r_2 = \frac{p_2}{p_1}$ is the asset's gross return.

²² We assume for simplicity that flows are insensitive to performance. In Internet Appendix I11, we extend the model to allow flows to depend on last period's fund return. Such performance-sensitive flows make the predictions of the model starker. Indeed, persistent flows continue to predict returns whereas transient flows do not, but the difference in predictive power between flow types is enhanced.

Market clearing. Arbitrageurs collectively buy $C_t = \int_i \theta_{t,i} A_{t,i}$ worth of the risky asset in period t , or C_t/p_t shares (where $C_0 = 0$). Since the asset supply is equal to one, market clearing implies that the price of the asset at $t < 3$ is given by $p_t = E_t(V) - s_t + C_t$. Thus, p_t decreases in noise traders' demand shock s_t , and increases in the capital C_t that arbitrageurs collectively commit to the trade.

The shocks to the fundamental value ($\varepsilon_t, t \geq 1$) and to period-2 flows (f_2) are the only sources of risk. Finally, to emphasize the dynamic properties of flows, we make the following assumption.

Assumption 2. f_1 is small and positive.

Internet Appendix I10 provides a formal statement of this assumption. Assuming f_1 is small relative to initial AUM (which are normalized to 1) ensures that outflows in period 2 do not exceed AUM. Assuming f_1 is small relative to mispricing implies that, in equilibrium, arbitrage capital is insufficient to eliminate the mispricing in periods 1 and 2; this highlights the role played by the persistence of flows, since the dynamics of flows would not matter if there were no shortage of capital in the future. Finally, with hedge-fund flows positive 70% of the time in our data, assuming f_1 is positive allows to focus on the empirically-relevant case of inflows (indeed, $f_1 > 0$ implies $f_1 + E(f_2|f_1) = 2qf_1 > 0$, i.e., flows are on average positive).

5.2 The equilibrium and its properties

Given that arbitrageurs do not participate in the risky asset's market in period 0, its price is $p_0 = v - s$. Proposition 1 describes the equilibrium in periods 1 and 2 for persistent and transient flows. For clarity, we label quantities in the case of transient flows with a prime (').

Proposition 1. Equilibrium Characterisation

The equilibrium is unique and characterised as follows:

1. Persistent flows ($q > 1/2$)

- At date 1, arbitrageurs hold no cash and invest all their AUM in the risky asset (i.e., $\theta_{1,i} = 1$ for all i). Hence the amount committed to the risky asset is $C_1 = \int_i A_{1,i} = 1 + f_1$. The asset's return and price are $r_1 = \frac{C_1 + \varepsilon_1}{v - s}$ and $p_1 = v + \varepsilon_1 - s + C_1$.
- At date 2, arbitrageurs hold no cash and invest all their AUM in the risky asset (i.e., $\theta_{2,i} = 1$ for all i). The asset's return and price are $r_2 = \frac{(1+f_1)f_2 + \varepsilon_2}{v + \varepsilon_1 - s}$ and $p_2 = v + \varepsilon_1 + \varepsilon_2 - s + C_1(1 + r_2 + f_2)$ where $f_2 = f_1$ if flows persist, and $f_2 = -f_1$ if they reverse.

2. Transient flows ($q \leq 1/2$)

- At date 1, arbitrageurs hold some cash, and invest in the risky asset the amount $C'_1 = (1 + f_1)(1 + E(f_2|f_1))$ where $0 < C'_1 < C_1$. The asset's return and price are $r'_1 = \frac{C'_1 + \varepsilon_1}{v - s}$ and $p'_1 = v + \varepsilon_1 - s + C'_1$.
- At date 2, arbitrageurs hold no cash (i.e., $\theta_{2,i} = 1$ for all i). The asset's return and price are $r'_2 = \frac{(1+f_1)(f_2 - E(f_2|f_1)) + \varepsilon_2}{v + \varepsilon_1 - s}$ and $p'_2 = v + \varepsilon_1 + \varepsilon_2 - s + C'_1(1 + r'_2 + f_2)$ where $f_2 = f_1$ if flows persist, and $f_2 = -f_1$ if they reverse.

See Internet Appendix I10 for all proofs. Arbitrageurs enter the market in period 1 and buy the underpriced asset, leading to a positive return r_1 or r'_1 . That return is larger, the more capital (C_1 or C'_1) they commit to the trade. How much they commit depends on whether flows are persistent, as we describe next.

In the case of persistent flows, arbitrageurs anticipate that the underpricing will be partially corrected in period 2 thanks to an inflow of capital into the arbitrage sector ($E(f_2|f_1) > 0$). The ensuing positive expected return ($E(r_2|f_1) > 0$) leads them to invest all their capital in the asset in period 1. If, instead, flows are transient, then they anticipate flows out of the sector in period 2 ($E(f_2|f_1) < 0$), and hence a smaller correction of underpricing. Given the asset's lower expected return, they invest less aggressively in period 1 ($C'_1 < C_1$), and hold a mix of the risky and riskless assets. For both persistent and transient flows, arbitrageurs are fully invested in the risky asset in period 2 because they expect then a strictly positive return ($E(r_3|f_2) > 0$, $E(r'_3|f_2) > 0$).

Note that AUM at $t = 2$ both determine the asset's return r_2 , and depend on it through the fund's return at $t = 2$ (which equals $\theta_{1,i} r_2$). This generates a feedback loop: if arbitrageurs are more optimistic about the risky asset's return in period 2, then they invest more in the asset in period 1. This increases their fund's return and so its AUM in period 2. With more capital committed to the risky asset in period 2, its price, and so its expected return, are even larger.

The next three propositions focus on the predictability of returns. We denote $(R_q^2)^{x \rightarrow y}$ the average R^2 from regressing a variable y (e.g., date-2 returns, r_2) on another variable x (e.g., date-1 flows, f_1). It represents the predictive power of x for y . Likewise, we denote $(R_q^2)^{x \rightarrow y|z}$ the average incremental R^2 , which captures the predictive power of x for y after conditioning on a third variable, z (e.g., period-1 returns, r_1). We start with the main implication of the model on the predictive power of flows for returns.

Proposition 2. Predictive Power of Flows for Returns

1. Persistent flows ($q > 1/2$) predict returns.

$$E(r_2|f_1) = \frac{1+f_1}{v+\varepsilon_1-s}E(f_2|f_1). \quad (R_q^2)^{f_1 \rightarrow r_2} > 0 \text{ and } (R_q^2)^{f_1 \rightarrow r_2|r_1} > 0.$$

2. *Transient flows* ($q \leq 1/2$) have no predictive power for returns.

$$E(r'_2|f_1) = 0 \text{ and } (R_q^2)^{f_1 \rightarrow r'_2} = (R_q^2)^{f_1 \rightarrow r'_2|r'_1} = 0.$$

Persistent flows have more predictive power for returns than do transient flows. This is true even when transient flows are more self-predictable. Formally, for $q > 1/2$ and $q' < 1 - q \leq 1/2$, $(R_q^2)^{f_1 \rightarrow r_2} > (R_{q'}^2)^{f_1 \rightarrow r'_2}$ and $(R_q^2)^{f_1 \rightarrow r_2|r_1} > (R_{q'}^2)^{f_1 \rightarrow r'_2|r'_1}$ even though $(R_q^2)^{f_1 \rightarrow f_2} < (R_{q'}^2)^{f_1 \rightarrow f_2}$.

Proposition 2 establishes that persistent flows predict returns whereas transient flows do not. This is true whether or not we condition on current returns, i.e., control for factor momentum. This finding is remarkable for three reasons. First, the sharp contrast in return dynamics between persistent and transient flows obtains despite the supply of arbitrage capital in period 1, as well as the size and timing of the demand shock, being identical in both cases. What drives the contrast is arbitrageurs' expectation of the *future* supply of *aggregate* capital in the arbitrage sector. Second, persistent flows have more predictive power than do transient flows *even* in the case in which transient flows are themselves more predictable than persistent flows. As an illustration, consider two probabilities, q and q' such that $q' > q > 1/2$. Then (persistent) flows with a continuation probability of q are less self-predictable than (transient) flows with a continuation probability of $1 - q'$ (i.e., $(R_q^2)^{f_1 \rightarrow f_2} < (R_{q'}^2)^{f_1 \rightarrow f_2}$). Yet, the former can predict returns more accurately ($(R_q^2)^{f_1 \rightarrow r_2} > (R_{q'}^2)^{f_1 \rightarrow r'_2}$ and $(R_q^2)^{f_1 \rightarrow r_2|r_1} > (R_{q'}^2)^{f_1 \rightarrow r'_2|r'_1}$). In an extreme case, suppose $q' = 0$: these flows reverse for sure but fail to predict returns. Finally, returns exhibit more predictability when flows are more persistent despite prices being *closer* to fundamentals. Thus, in the presence of frictions, higher return predictability need not be a sign of a more inefficient market. This result underscores the importance of understanding the implications of flow persistence for market efficiency.

Intuitively, arbitrageurs start correcting mispricing as soon as they enter the market in period 1. How aggressive they are depends on their anticipation of future mispricing corrections, which in turn depends on how much capital they expect will flow into the arbitrage sector. When flows are persistent, arbitrageurs expect a price appreciation in period 2 so strong that it cannot be fully priced in in period 1 given the limited supply of arbitrage capital in period 1. This generates return predictability from flows. When flows are transient, in contrast, arbitrageurs anticipate a moderate price appreciation in period 2, since arbitrage capital will then be

scarce. In this case, the capital available in period 1 suffices to exhaust the trading opportunity so that the asset's expected return is driven down to zero and there is no predictability.

Again, the reason returns are predictable in the case of persistent flows is that investors cannot push the period-1 price up to its period-2 expected level (or equivalently, push expected returns down to 0) due to a lack of capital. Importantly, what matters is not the supply of capital in period 1—it's the same f_1 under persistent and transient mispricing—but the supply relative to the expected profitability of the investment. This profitability in turn depends on the period-2 price, which itself depends on capital availability in period 2. In other words, what determines return predictability is not capital scarcity in period 1 but *its scarcity in period 1 relative to its expected scarcity in period 2*. This observation underscores the importance of the dynamic properties of flows.

The next proposition characterizes the contemporaneous relation between flows and returns. This proposition, unlike all others, requires that we specify the distribution of period-1 flows, f_1 . We assume that they are uniformly distributed: $f_1 \sim \mathcal{U}[0,1]$.

Proposition 3. Contemporaneous Relation between Flows and Returns

Assume that period-1 flows are uniformly distributed: $f_1 \sim \mathcal{U}[0,1]$. Persistent flows display a larger coefficient in contemporaneous flow-return regressions than do transient flows. They also display more explanatory power for returns as long as transient flows are not almost perfectly predictable (i.e., their probability of persisting, q' , is larger than $1/122 \approx 0.8\%$). Formally, $(R_q^2)^{f_1 \rightarrow r_1} > (R_{q'}^2)^{f_1 \rightarrow r'_1}$ for any $q > 1/2$ (persistent) and $1/2 > q' > 1/122$ (transient).

Proposition 3 implies that the coefficient from regressing returns on contemporaneous flows is larger when flows are persistent than when they are transient. Moreover, the explanatory power is greater provided transient flows are not almost perfectly predictable. Intuitively, capital in period 1 is more constrained when flows are more persistent. Indeed, arbitrageurs invest all their AUM in the risky asset when flows are persistent, but only part of it when flows are transient. This makes period-1 returns more responsive to contemporaneous flows in the case of persistent flows.

Finally, we examine the self-predictability of factor returns (i.e., factor momentum) and how it relates to the predictive power of flows for returns.

Proposition 4. Returns' Self Predictability

1. Persistent flows ($q > 1/2$).

Current returns predict future returns. Formally, $(R_q^2)^{r_1 \rightarrow r_2} > 0$. Returns display a mix of momentum, caused by flows' continuations, and reversals, caused by fundamental shocks. Factor momentum arises as long as mispricing is moderate or fundamental shocks are not too large.

2. Transient flows ($q \leq 1/2$).

Current returns have no predictive power for future returns: $(R_q^2)^{r'_1 \rightarrow r'_2} = 0$.

Only if flows are persistent do current returns predict future returns, or does factor momentum arise (i.e., return continuations dominate reversals) under plausible conditions. If flows are transient, then returns are unpredictable from past returns, even though flows themselves are predictable. In addition, the predictive power of returns in the case of persistent flows derives entirely from the predictability of flows: r_1 is informative about contemporaneous flows, f_1 , which predict future flows f_2 , which in turn relate to contemporaneous returns, r_2 .²³

5.3 Predictions

The model's two main implications, stated in Propositions 2 and 3, conform to our empirical findings. The first is that persistent, but not transient, flows predict future returns. The second is that persistent but not transient flows are closely related to ("explain") returns contemporaneously.²⁴ Both predictions hold despite transient flows being equally or even more predictable than persistent flows in the data.

Prediction 1. Persistent, but not transient, flows predict returns.

Prediction 2. Persistent flows have more explanatory power for returns and yield a larger regression coefficient in contemporaneous regressions than do transient flows.

The model yields three additional predictions that can be brought to the data to evaluate the underlying mechanism. The model's third prediction, which follows from Proposition 1, is that arbitrageurs trade the mispriced asset more aggressively when flows are persistent than when they are transient. Indeed, arbitrageurs commit more capital in period 1 to the underpriced asset and hold less cash when flows are more persistent. Cash in the model should be interpreted broadly as representing any liquid assets; more liquid assets indeed are less likely to be mispriced (e.g., Aragon 2007).

Prediction 3. Arbitrageurs trade the mispriced asset more aggressively and hold less cash (and liquid assets) when flows are persistent than when they are transient.

²³ Returns have no predictive power above and beyond that of flows. To illustrate this point, observe that if flows become unpredictable (i.e., $q = 1/2$), then returns too lose their predictive power (i.e., $Cov(r_1, r_2) = 0$, so $E(r_2|r_1) = E(r_2|f_1) = 0$ and $(R_{1/2}^2)^{r_1 \rightarrow r_2} = 0$).

²⁴ In the data, transient flows are far from perfectly predictable. Hence the condition in Proposition 3 is satisfied, leading to Prediction 2.

The fourth testable prediction is that (persistent) flows have both more predictive and more explanatory power for returns when arbitrage capital is more scarce. For example, when the borrowing constraint is removed, the price converges to the fundamental in period 1, leading to returns of $(s + \varepsilon_1)/(v - s)$ in period 1 and $\varepsilon_2/(v + \varepsilon_1)$ in period 2, both of which are independent of flows (and unpredictable). This prediction holds in the time series as well as in the cross-section. To see how it applies across funds, suppose the market comprises two types of funds and two risky assets. The unlevered (levered) fund is managed by constrained (unconstrained) arbitrageurs who cannot (can) borrow, and specializes in risky asset C (U). The two assets are ex-ante identical: they have the same fundamental value v and mispricing s , and are subject to identically distributed shocks ε_τ^C and ε_τ^U ($\tau = 1, 2, 3$), with parameters conforming to Assumptions 1 and 2. The model implies that flows to constrained arbitrageurs predict the return of asset C , whereas flows to unconstrained arbitrageurs have no predictive power for the return of asset U . Indeed, the price of the latter adjusts to its fundamental value in each period, $p_t^U = E_t(V^U) = v + \sum_{\tau=1}^t \varepsilon_\tau^U$, and so its returns are not predictable. As a result, the average return across the two assets can be predicted using either aggregate flows or the flows of constrained arbitrageurs; but it cannot be predicted using the flows of unconstrained arbitrageurs. Hence only flows to unlevered fund (constrained arbitrageurs) display predictive power.

Prediction 4. In the time series, returns are more sensitive to persistent flows when arbitrage capital is scarcer. In the cross-section, persistent flows of constrained arbitrageurs (unlevered fund) predict the returns of assets traded by constrained arbitrageurs (as well as the returns of any portfolio containing those assets), whereas persistent flows of unconstrained arbitrageurs (levered funds) do not predict the returns of assets traded by unconstrained arbitrageurs.

Finally, Proposition 4 predicts that, when flows are *persistent*, factor returns are *self*-predictable. Thus, the combination of leverage constraints with flow persistence generates factor momentum. This mechanism differs from that proposed by Ehsani and Linnainmaa (2022a), which relies on sentiment (see Internet Appendix I9).

Prediction 5. Persistent flows lead to factor return self-predictability.

In closing, we note that, while our model fits well the behavior of hedge funds who are the archetypical arbitrageurs, and in the data, appear to correct mispricing, it can also shed light on the behavior of mutual funds. To do so, we interpret mutual funds as “arbitrageurs” who wrongly believe the asset to be overpriced. To see that, assume that mutual funds believe that the stock’s fundamental value is not v but $v^* < v - s$ (or,

equivalently, they correctly view the stock's fundamental value as v but wrongly believe that mispricing will worsen, i.e., the demand shock will increase to $s^* > s$ rather than disappear in period 3). In that scenario, they short, rather than buy, the stock, thereby exacerbating underpricing. As a result, asset returns and mutual fund flows are *negatively* related, as in the data. Indeed, capital-constrained mutual funds short more shares of the asset when they receive more flows, which, in turn, reduces its price and leads to negative (contemporaneous) returns. Importantly, the main implications of the model continue to hold. In particular, mutual funds sell the asset aggressively in period 1 if, and only if, they expect flows to persist (so they expect the price to decrease further in period 2). This leads to persistent, but not transient, flows predicting future returns (as in Proposition 2) and explaining contemporaneous returns (as in Proposition 3). That being said, mutual-fund flows' predictive power for returns is counteracted by their cross-autocorrelation with future hedge-fund flows as noted in Section 3.2. Therefore, their behavior is more complex than hedge funds' and thus goes beyond the scope of this paper, which focuses on arbitrage capital.

6. Further evidence supporting the economic mechanism

In Sections 2 and 4, we provided evidence consistent with Predictions 1 and 2 of the model, respectively. In Internet Appendix I7 and Table I7-1, we present additional evidence supporting the central role of arbitrageurs; specifically, we document that persistent flows have more predictive power for the stocks in the SYY portfolio that are less costly to short, as well as for longer-horizon and lower-turnover factor returns. In this section, we examine Predictions 3-5 (Section 6.1-6.3).

6.1 How do managers slow down flows?

According to Prediction 3, (i) funds trade more aggressively on anomalies, and (ii) exhibit a weaker tendency to park flows into liquid assets, when those flows are persistent compared to when they are transient. We evaluate next both implications.

We use quarterly holdings data retrieved from 13F filings to estimate funds' trading intensity in anomaly stocks.²⁵ A positive (negative) intensity signifies that funds buy (sell) more—or sell (buy) less—long-leg stocks

²⁵ We follow Dong et al. (2024) to compute anomaly trading intensity in three steps. First, we measure hedge (mutual) funds' net buying in any given stock by aggregating the number of shares held in the stock by hedge (mutual) funds at the end of the quarter, dividing by the number of shares outstanding, and computing relative quarterly changes. Second, for each anomaly, we average hedge (mutual) funds' net buying over all stocks comprising the anomaly's long leg, and proceed similarly for the short leg. Here, we identify long- and short-leg stocks using data from the end of the previous quarter. Finally, we define the anomaly's trading intensity as the difference between the average net buying of the long and short legs, normalized by the standard deviation of this difference.

than short-leg stocks, and in doing so, contribute to correcting (exacerbating) mispricing. Quarterly regressions reported in Panel A of Table 7 reveal a significant effect of low-, but not high-frequency, flows on hedge (mutual) funds' trading intensity, which is positive (negative). A 1-SD increase in low-frequency flows leads to a 0.14-SD (0.11-SD) increase in hedge (mutual) funds' trading against (with) anomalies.^{26,27} These findings, in addition to supporting Prediction 3, provide *direct* evidence that hedge (mutual) funds deploy flows to correct (exacerbate) mispricing, as claimed in Sections 3.2 and 4.3.

Table 7: Funds' Trades

The table reports results from quarterly regressions of funds' trading activity on the low- and high-frequency components of fund flows. In Panel A, the dependent variable is funds' trading intensity in anomaly stocks, and in Panel B, their tendency to invest in illiquid stocks. We use holdings data retrieved from quarterly 13F filings to measure funds' trading activity. We estimate funds' trading intensity in anomaly stocks (Panel A) as follows: first, we measure hedge (respectively, mutual) funds' net buying in any given stock by aggregating the number of shares held by hedge (mutual) funds at the end of the quarter, dividing by the number of shares outstanding, and computing relative quarterly changes; second, for each anomaly, we average hedge (mutual) funds' net buying over all stocks comprising the anomaly's long leg and short leg; finally, we define the anomaly's trading intensity as the difference between the average net buying of the long legs and that of the short legs, normalized by the standard deviation of this difference. We measure funds' tendency to invest in illiquid stocks (Panel B) as follows: first, we sort stocks into deciles according to their illiquidity measured over the previous quarter (based on the Amihud illiquidity ratio or bid-ask spread), and assign them a score from one (most liquid) to ten (least liquid); then we calculate funds' tendency to invest in illiquid stocks as the difference in the average illiquidity between the stocks they buy and those they sell. Dependent variables in the regressions are standardized to have mean zero and unit standard deviation. The control variables include MKTRF, aggregate Amihud measure, aggregate turnover, HML and SMB. *t*-statistics are calculated based on Newey-West standard errors with 13 lags. The sample period is 1994–2020.

Panel A: Anomaly Trading Intensity

	(1)		(2)		(3)		(4)	
Funds Type	HF				MF			
FLOW	5.011	[2.11]			-8.542	[-2.32]		
FLOW-LOW			4.139	[2.16]			-8.526	[-1.84]
FLOW-HIGH			7.742	[1.41]			-8.610	[-1.06]
Controls	Yes		Yes		Yes		Yes	
N	107		107		107		107	
R ²	3.0%		3.2%		10.3%		10.3%	

Panel B: Liquidity of Funds' Trades

Liquidity Measure Funds Type	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Bid-Ask Spread				Amihud Illiquidity			
	HF		MF		HF		MF	
FLOW	12.352		34.005		11.800		21.285	
	[3.85]		[1.98]		[3.40]		[2.54]	
FLOW-LOW		26.435		64.227		26.191		48.434
		[4.36]		[2.09]		[4.16]		[3.24]
FLOW-HIGH		1.459		4.759		0.669		-4.987
		[0.49]		[0.33]		[0.21]		[-0.71]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	323	323	323	323	323	323	323	323
R ²	12.7%	15.9%	28.5%	30.2%	13.3%	16.6%	13.5%	15.0%

To evaluate the second implication of Prediction 3, we sort stocks into deciles according to their illiquidity

²⁶ E.g., for HF-LOW, $0.14 = 4.14 \times 3.37\%$. The standard deviations of quarterly HF, MF, HF-LOW, HF-HIGH, MF-LOW, and MF-HIGH are 3.66%, 1.47%, 3.37%, 1.54%, 1.34%, and 0.48%, respectively. Recall that the anomaly trading intensity is normalized.

²⁷ In Table I8-7 of the Internet Appendix, we re-estimate the results of Panel A, using NINV anomalies. While in Panel A, the point estimate of HF-HIGH is larger, albeit insignificant, than that of HF-LOW, Table I8-7 shows that for NINV anomalies, the point estimate for high-frequency hedge fund is not only insignificant but also lower in magnitude than that of low-frequency.

over the previous month, assigning them a score from one (most liquid) to ten (most illiquid). We compute two scores, one based on the Amihud illiquidity ratio and the other on the bid-ask spread. We measure funds' tendency to invest in illiquid stocks as the difference in the average score between the stocks they buy and those they sell, normalized by the standard deviation of this difference. Thus, a positive (negative) value for this difference points to funds net-buying illiquid (liquid) stocks. Regressions of funds' tendency to buy illiquid stocks on decomposed flows, reported in Panel B of Table 7, yield significant estimates at low frequency only; these are positive, implying that managers buy more illiquid stocks upon receiving lower-frequency flows. Based on the Amihud ratio for instance, a 1-SD increase in low-frequency hedge- (mutual-) fund flows results in a 0.88-SD (0.65-SD) increase in funds' tendency to buy illiquid stock. Hence, low-frequency flows are less likely to be held in liquid stocks than high-frequency flows, especially for hedge funds, again consistent with Prediction 3.

6.2 Why do managers slow down flows? The role of constraints

Table 8: The Impact of Leverage Constraints

The table examines the impact of leverage on the low-frequency tilt. The dependent variables are the low- and high-frequency components of SYY factor returns. In Panel A, the main independent variables are the low- and high-frequency components of fund flows, and their interactions with LEV, which represents funding cost measured using the TED Spread. LEV is a quintile score scaled from zero (low funding cost or risk aversion) to one (high funding cost or risk aversion). For Panel B, we divide hedge funds into two groups according to their use of leverage, HFUnLev (unlevered funds) and HFLev (levered funds). Then, fund flows are aggregated separately over each group. All regressions include MKTRF, aggregate Amihud measure, aggregate turnover, HML, and SMB, as control variables. *t*-statistics are calculated based on Newey-West standard errors with 13 lags. The sample period is 1994–2020.

Panel A: Funding Cost

	(1)		(2)	
	FREQ = LOW		FREQ = HIGH	
HF-FREQ	0.300	[1.48]	0.585	[1.83]
HF-FREQ × LEV	0.802	[2.63]	-0.718	[-1.32]
MF-FREQ	-3.304	[-5.75]	0.503	[0.45]
MF-FREQ × LEV	-0.583	[-0.62]	-2.651	[-1.26]
N	324		324	
R ²	35.7%		40.7%	

Panel B: Hedge-Fund Leverage

	(1)		(2)	
	FREQ = LOW		FREQ = HIGH	
HFUnlev-FREQ	0.918	[2.21]	-0.255	[-0.61]
HFLev-FREQ	-0.149	[-0.34]	0.384	[1.43]
MF-FREQ	-3.672	[-4.13]	-2.558	[-2.93]
N	324		324	
R ²	33.4%		33.5%	

Prediction 4 states that the effect of persistent, but not transient, flows on factor returns is more pronounced when capital is more restricted, thereby strengthening the low-frequency tilt. To evaluate this prediction, we consider variations in capital accessibility over time and across hedge funds—we leave out mutual funds as they do not

differ much along that dimension. The time-series analysis, reported in Table 8, Panel A, employs the TED spread as a proxy for funding costs (which make leverage more expensive). We interact flows with LEV, a quintile score scaled from zero to one, where a higher score indicates a wider TED spread (higher cost). As predicted, the estimates for $\text{HF-LOW} \times \text{LEV}$ are positive, large and significant, whereas those for $\text{HF-HIGH} \times \text{LEV}$ are insignificant. In Column (1), for example, the magnitude of the effect at low frequency almost quadruples from 0.300 in the lowest TED-spread quintile to 1.102 ($=0.802+0.300$) in the highest.

Examining the cross-section of funds, we conduct two types of analyses, one at the fund level, and the other at the institution level. Starting with the fund-level analysis, we note that unlevered hedge funds are likely to face more restrictions on borrowing than do levered funds (e.g., Adrian and Shin 2013), and hence should display a more pronounced effect of persistent flows. Each month, we classify hedge funds into two groups using Lipper TASS data, which provide a range of fund-level characteristics, including the use of leverage (as an indicator variable): funds that use leverage (HFLev) and those that do not (HFUnLev). We refer to the former funds as “unconstrained” and to the latter as “constrained”. Then we calculate flows separately for each group, decompose each series across frequencies, and regress factor returns on the resulting components. Panel B of Table 8 shows that only HFUnLev exhibits a significantly positive coefficient estimate at low frequency, consistent with our prediction.

Next, we turn to an institution-level analysis which allows us to compare the holdings—extracted from 13F filings—of more levered hedge-fund institutions to those of less levered institutions. Importantly, 13F holdings are reported at the institution level, not the fund level, and a 13F hedge-fund institution typically manages multiple funds with varying leverage profiles. To obtain an institution-wide measure of leverage, we start from the categorization of hedge funds in Table 8, based Lipper TASS data, that classified funds as unlevered and levered. Then, we establish a mapping between TASS data and 13F filings through a combination of fuzzy name and manual matching. Finally, we calculate a leverage ratio for each 13F institution as the proportion of AUM attributed to funds employing leverage relative to the total AUM of the institution. We refer to 13F institutions with a leverage ratio below 0.5 as (relatively) constrained, and those with a ratio above as unconstrained. In other words, for a constrained institution, most assets (>50% of its AUM) are from unlevered funds, while for an unconstrained institution, most assets are from levered funds. This procedure yields a total of 559 unique hedge-fund institutions, with about a third classified as constrained.

At the onset, we note that, for return predictability to arise from capital constraints—as the fund-level results in Panel B of Table 8 suggest—constrained and unconstrained institutions should follow different investment strategies (e.g., invest in different stocks or invest in similar stocks but with a different intensity). If they did not, unconstrained institutions would simply undo the effect of the constraints faced by constrained institutions. Accordingly, the first step of the holdings analysis is to check whether these two types of institutions invest in largely distinct stock universes. To do so, for each stock in the SYR portfolio, we calculate the average holding in the stock (as a percentage of institutions' AUM), separately for constrained and unconstrained institutions:

$$H_{s,t}^C = \frac{1}{N_t^C} \sum_{i=1}^{N_t^C} \frac{\$Holding_{s,i,t}}{AUM_{i,t}} \quad \text{and} \quad H_{s,t}^U = \frac{1}{N_t^U} \sum_{i=1}^{N_t^U} \frac{\$Holding_{s,i,t}}{AUM_{i,t}},$$

where $H_{s,t}^C$ ($H_{s,t}^U$) denotes the average holding in stock s among constrained (unconstrained) institutions in quarter t , $AUM_{i,t}$ is the AUM of fund i in quarter t , $\$Holding_{s,i,t}$ is the value of fund i 's position in stock s in quarter t , and N_t^C (N_t^U) is the number of constrained (unconstrained) institutions in quarter t . Next, we calculate the absolute value of the difference between the average holdings, in the stock, of constrained institutions and the average of unconstrained institutions: $|H_{s,t}^C - H_{s,t}^U|$. Finally, we calculate the mean and median of those absolute differences across all stocks in the SYR portfolio. These statistics measure the degree to which constrained institutions invest in a different set of SYR stocks from unconstrained institutions. Indeed, if both types of institution invest in similar stocks, then the absolute difference should not be distinguishable from zero.²⁸ To assess significance, we compare the mean (median) of absolute differences to the mean (median) of the average holdings across all institutions, $H_{s,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{\$Holding_{s,i,t}}{AUM_{i,t}}$, where N_t ($= N_t^C + N_t^U$) denotes the total number of institutions in quarter t .²⁹

Table 9: Investing Universe of Constrained and Unconstrained Institutions

This table compares the holdings in SYR stocks of constrained hedge-fund institutions to those of unconstrained institutions. $H_{s,t}^C$ ($H_{s,t}^U$) denotes the average holding in stock s among constrained (unconstrained) institutions in quarter t . Holding in stock s by institution i is calculated from the dollar value of holding in stock s by institution i divided total AUM of i . Panel A provides the summary statistics of

²⁸ Due to the limitations of 13F data (recall that unconstrained institutions may comprise some levered funds, while constrained institutions may comprise some unlevered funds), we cannot distinguish institutions investing in entirely different stocks (extensive margin) from institutions investing in the same stocks but with markedly different holdings relative to their AUM (intensive margin). We still consider the latter as evidence of segmentation, since it suggests that constrained and unconstrained institutions invest differently in SYR stocks, i.e., with significantly different portfolio weights.

²⁹ By construction, the mean and median of $|H_{s,t}^C - H_{s,t}^U|$ are significantly positive. We compare them to the mean and median holdings to account for measurement error. Indeed, under the null of no difference between the holdings of constrained and unconstrained institutions, the absolute difference might still be significantly positive if holdings are reported imprecisely.

average holdings by the types of institutions and Panel B reports the hypothesis testing results. The *t*-statistics are based on standard errors clustered at the firm level. The sample period is 1994–2020.

Panel A: Summary Stat - Average Portfolio Weight on SYX Stocks

	N	Mean	Median	Std Dev
Pooled	280846	0.36%	0.06%	1.41%
Constrained (H^C)	140423	0.35%	0.02%	1.45%
Unconstrained (H^U)	140423	0.37%	0.11%	1.37%
Distance ($ H^C - H^U $)	140423	0.47%	0.12%	1.81%

Panel B: Hypothesis Testing

Hypothesis	Test	Statistics	p value
Mean of $ H^C - H^U > \text{Sample Mean}$	<i>t</i> Test	8.42	<.0001
Median of $ H^C - H^U > \text{Sample Median}$	Sign Test	16711.50	<.0001
Median of $ H^C - H^U > \text{Sample Median}$	Signed Rank Test	2.83×10^9	<.0001

Table 9 provides the results. Panel A reports the summary statistics for $H_{s,t}^C$ and $H_{s,t}^U$, and their absolute difference. The panel shows that the mean (median) absolute difference is economically large and statistically significant. Panel B presents the hypothesis testing results. These tests reject, at any conventional level, the null hypothesis that the mean (median) of the absolute differences is equal to, or smaller than, the sample mean (median) of the average holdings. In addition, the economic magnitudes are large: the mean of the absolute differences between the two types of institutions is 31% ($=0.47\%/0.36\%-1$) larger than the mean of their average holdings. Likewise, the median of the absolute differences is 101% ($=0.12\%/0.06\%-1$) larger than the median of the average holdings. These findings demonstrate that constrained and unconstrained institutions select largely distinct SYX stocks for their portfolios.

Table 10: Constrained vs. Unconstrained Factors

This table compares the predictability of two distinct SYX factors: a constrained SYX factor and an unconstrained SYX factor. Specifically, we estimate, for each stock and institution type, a trading intensity defined as the change in the dollar value of the stock's holdings divided by the stock's market capitalization. Then, we construct each leg of the constrained and unconstrained SYX factors as follows. The long leg of the constrained (unconstrained) SYX factor consists of stocks within the top decile of anomaly scores that fall into the top 30% of trading intensity by constrained (unconstrained) institutions. Conversely, the short leg consists of stocks within the bottom anomaly score decile that belong to the bottom 30% of trading intensity by constrained (unconstrained) institutions. Panel A reports the OOS predictability results using the MA(6) of flows to entire hedge funds, while Panel B provides the OOS results of predicting the total SYX factor using flows to constrained and unconstrained funds. Panel C reports the OOS results for predicting the constrained (unconstrained) SYX factor using flows to constrained (unconstrained) funds. The dependent variables in Panels A, B, and C are multi-period factor returns, $r_{t,t+j}$, where $j=0, 2$, and 5 . Thus, the forecasting horizons for these multi-period returns are 1, 2, and 6. Panel D reports the contemporaneous factor-flow relation for the constrained (unconstrained) factor and flows. The sample period is 1994–2020.

Panel A: OOS R2 - Predicting Constrained/Unconstrained Factors Based On Total HF-MA(6)

		(1)	(2)	(3)	(4)	(5)	(6)
Factor Type		Constrained Factor			Unconstrained Factor		
Forecasting Horizon (Months)		1	3	6	1	3	6
R2(%)	Point Estimate	1.18	4.56***	1.62**	-1.74	-4.03	-4.93
	Lower Bound (10%)	-0.01	2.37	0.54	-3.52	-7.99	-9.00
	Lower Bound (5%)	-0.34	1.74	0.24	-4.02	-9.12	-10.16

Panel B: OOS R2 - Predicting Total SY Y Factor Using Flows to Constrained/Unconstrained Funds

		(1)	(2)	(3)	(4)	(5)	(6)
Hedge-Fund Type		Constrained Flows			Unconstrained Flows		
Forecasting Horizon (Months)		1	3	6	1	3	6
R2(%)	Point Estimate	7.59**	12.36***	16.76***	3.48**	3.05**	1.09**
	Lower Bound (10%)	3.23	7.05	10.49	1.22	1.26	0.39
	Lower Bound (5%)	1.98	5.53	8.71	0.57	0.75	0.19

Panel C: OOS R2 - Constrained vs. Unconstrained Pairs

		(1)	(2)	(3)	(4)	(5)	(6)
Factor (Hedge-Fund) Type		Constrained Factor and Flows			Unconstrained Factor and Flows		
Forecasting Horizon (Months)		1	3	6	1	3	6
R2(%)	Point Estimate	2.54**	9.91***	15.68***	-0.35	0.30	-5.13
	Lower Bound (10%)	0.66	5.71	10.50	-3.73	-0.07	-8.88
	Lower Bound (5%)	0.13	4.52	9.02	-4.69	-0.17	-9.95

Panel D: Contemporaneous Factor-Flow Relations

		(1)	(2)	(3)	(4)
Factor (Fund) Type		Constrained Factor and Flows		Unconstrained Factor and Flows	
		FREQ = LOW	FREQ = HIGH	FREQ = LOW	FREQ = HIGH
HF-FREQ		0.823	0.324	0.447	1.256
		[3.71]	[0.66]	[1.82]	[4.16]
MF-FREQ		-3.826	-2.834	-3.349	-4.652
		[-4.95]	[-2.16]	[-3.88]	[-3.01]
Controls	Yes	Yes	Yes	Yes	Yes
N		322	322	322	322
R ²		33.0%	22.8%	30.7%	22.3%

We can now proceed to the final part of the holdings analysis. According to Prediction 4, persistent flows of constrained arbitrageurs predict the returns of assets traded by constrained arbitrageurs, whereas persistent flows of unconstrained arbitrageurs do not predict the returns of assets traded by unconstrained arbitrageurs. Accordingly, we construct two distinct SY Y factors: a constrained SY Y factor, comprising stocks that are predominantly traded by constrained institutions, and an unconstrained SY Y factor comprising stocks that are predominantly traded by unconstrained institutions. Specifically, we estimate, for each stock and institution type, a trading intensity defined as the change in the dollar value of the stock's holdings divided by the stock's market capitalization. Then, we construct each leg of the constrained and unconstrained SY Y factors as follows. The long leg of the constrained (unconstrained) SY Y factor consists of stocks within the top decile of anomaly scores that fall into the top 30% of trading intensity by constrained (unconstrained) institutions. Conversely, the short leg consists of stocks within the bottom anomaly score decile that belong to the bottom 30% of trading intensity by constrained (unconstrained) institutions.

We conduct the same analysis on those two distinct SY Y factors as in Table 2 (predictive OOS regressions) and Table 4 (contemporaneous factor-flow relation). The results, presented in Table 10, are fourfold. First, Panel A shows that the returns of SY Y stocks predominantly traded by constrained institutions are predictable through flows, with OOS R^2 s ranging from 1.18% to 4.56%, while the returns of SY Y stocks predominantly traded by unconstrained institutions display little predictability. Put differently, the OOS predictability of the (total) SY Y factor is attributable to the constrained SY Y factor. Second, Panel B reports flows directed to unlevered funds display considerably (from 2 to 10 times) more predictive power for (total) SY Y returns than do flows towards levered funds. Third, combining these two findings, we show in Panel C that persistent flows to unlevered funds strongly predict constrained factor returns OOS, whereas flows to levered funds do not predict unconstrained factor returns. The OOS R^2 for the former ranges from 2.54% to 15.68%.

Finally, we estimate the *contemporaneous* factor-flow relation for the pairs of flow and factor series decomposed across frequencies. Panel D reveals that the positive relation between the low-frequency flows and low-frequency factor returns is markedly stronger for the constrained factor and the unlevered flows compared to that between the unconstrained factor and levered flows. Specifically, the coefficient estimates and t -statistics for the constrained factors and unlevered flows are about twice as large as those for the unconstrained factors and levered flows. The effect of high-frequency flows is also strikingly different across pairs. Whereas flows fail to explain returns at high frequency for the constrained factor and unlevered fund flows—in line with the low-frequency tilt, they possess strong explanatory power at high frequency for the unconstrained factor and levered flows. In fact, those exhibit no such tilt.

All in all, the results in this section highlight the central role played by constraints faced by managers in driving the predictive power of persistent flows.

6.3 Factor returns' self-predictability and factor momentum

We examine the final prediction of our model which pertains to factors' self-predictability. Although monthly factor returns may contain both momentum and reversal components, their combined effect in the data often results in momentum. Empirically, factor momentum is observed in-sample rather than OOS (see Haddad et al (2020, Table 5), as well as Panel E of our Table 3). Our empirical results contribute to understanding in-sample factor momentum in two ways.

First, in Panel B of Table 5, we examined the contribution of the autocorrelation of flows to the autocorrelation of factor returns over the low and high frequency bands (as we did in Section 4.2 for their variance). This analysis helps determine the extent to which factor momentum (reversal) is attributable to flow momentum (reversal). As demonstrated in Section 3.1, this contribution is determined by the R^2 of the contemporaneous regressions displayed in Table 4. The results indicate that, at low frequency, the contributions of hedge- and mutual-fund flows to factors' autocorrelation amount to, respectively, 13.6% and 42.0% (Column (6)). Since flows and factor returns are positively autocorrelated at low frequency, and low-frequency returns represent a substantial component of total returns (Column (5) of Panel A), this finding implies that persistent flows, i.e., flow momentum, explain a sizable portion of factor momentum. At high frequency, in contrast, fund flows contribute little to returns' autocorrelation. Therefore, transient flows, i.e., flow reversals, largely do not explain high-frequency reversals in factor returns. These results are again a manifestation of the low-frequency tilt. They lend support to Prediction 5 of the model.

Figure 1: Attribution of Factor's Self Predictability to Factor Return Components

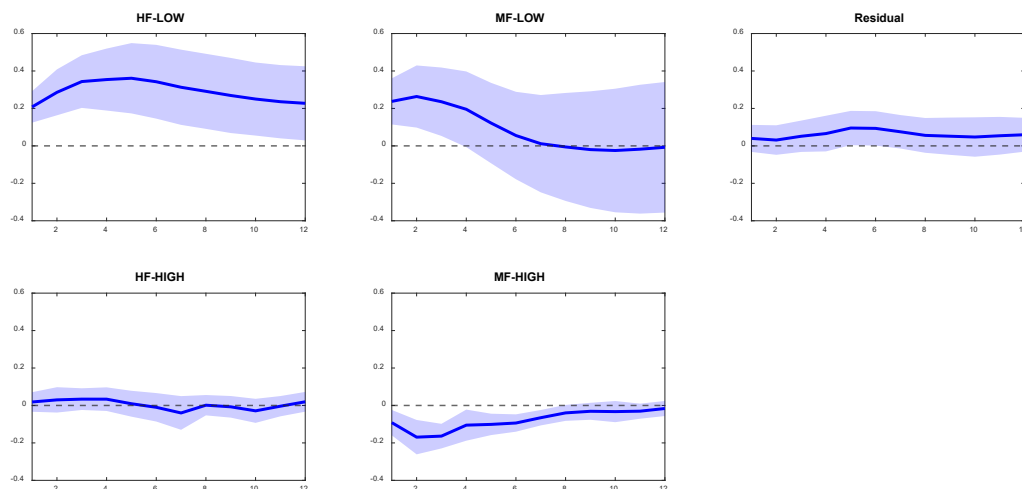
This figure shows the regression coefficients of the cumulative raw SYY factor return over the next j months, $r_{t+1,t+j}$ ($j = 1$ to 12), on decomposed monthly SYY factor return components, $\hat{r}_t$. Monthly SYY factor return, r_t , is decomposed into the components explained by low-frequency (high-frequency) hedge-fund (mutual-fund) flows. Specifically, we estimate the following regression:

$$r_t = \gamma_0 + \gamma_1 f_t^K + \gamma_2' F_t + \gamma_3' X_t + \varepsilon_t,$$

where f_t^K (for $K = Low$ or $High$) denotes the K -frequency component of hedge- (mutual-) fund flows, F_t is a vector of high- and low-frequency fund flows except for f_t^K , and X_t is a set of control variables, including the Fama-French three factors, the aggregate turnover, and the aggregate Amihud illiquidity. Then, using the fitted value based on f_t^K , $\hat{r}_t = \hat{\gamma}_1 f_t^K$, and the residual of the regression, ε_t , we estimate the following regression:

$$r_{t+1,j} = \alpha + \rho_1 \hat{r}_t + \rho_2 \varepsilon_t + \delta_1' F_t + \delta_2' X_t + v_{t+1}^K.$$

The x-axis of each panel is the return horizon, j , and the y-axis reports to the regression coefficient, ρ_1 . The blue line plots the coefficient for each period and the shaded area indicates the confidence interval at the 10% significance level. t -statistics are calculated based on Newey-West standard errors with 13 lags. All variables are standardized to have mean zero and unit standard deviation. The sample period is 1994–2020.



Second, to further evaluate the above conclusion, we examine how much of returns' self-predictability can

be attributed to the components of factor returns associated with low- and high-frequency flows. To do so, we proceed in two stages. First, we run a contemporaneous regression of factor returns on the low- and high-frequency components of flows along with control variables as in Table 4. Then, we estimate the return explained by low- and high-frequency flows as the coefficient estimate for those flows multiplied by the flows. In the second stage, we regress future factor returns cumulated over horizons ranging from one to 12 months on the values explained by flow components, the control variables and residuals from the first-stage regression. Figure 2 reports the coefficient estimates from those predictive regressions. Variables are standardized to facilitate comparisons. The figure indicates that the values explained by low-frequency flows positively predict factor returns, consistent with the presence of factor momentum. For hedge funds in particular, the coefficients are significant over horizons up to 12 months, peaking at about 0.4. This implies that 40% of the flow-explained factor return realizations continue into the future, thus leading to factor momentum. In contrast, the values explained by high-frequency flows display no predictive power. The only exception is for high-frequency mutual-fund flows over seven-month horizons, for which the negative and small coefficient estimates imply a weak tendency for factor returns linked to high-frequency mutual-fund flows to reverse. Finally, the first-stage regression residual does not significantly predict returns. All in all, our results suggest that factor momentum is, to a large extent, driven by persistent flows, while flow reversals matter modestly for factor reversals.

7. Accounting for the endogeneity of flows: Granular Instrument Variables

We deal with the endogeneity of flows with respect to factor returns. Before proceeding to formal tests, we note that the contrasting patterns we report across fund types should alleviate endogeneity concerns. Indeed, if the flow-return relationship is driven by an omitted variable affecting flows and factor returns simultaneously, then that variable needs to exert, at each frequency, (a) opposite influences on hedge- and mutual-fund flows (given the opposite signs we find in Table 4 for their effects on returns), and (b) no influence on passive mutual-fund flows (see Table I8-6 in the Internet Appendix I8). It is difficult to entertain (a), and even more (a) and (b) simultaneously, considering flows' high correlation (e.g., in Table I1-2 in the Internet Appendix I1, the correlations of mutual-fund flows with passive- and hedge-fund flows range from 0.54 to 0.57 at low frequency). Likewise, for reverse causality to hold, factor returns need to affect hedge- and mutual-fund flows in opposite directions while not affecting passive fund flows. In addition, the theory proposed in Section 5, supported by the

evidence in Section 6, reinforces the plausibility of a causal effect. Specifically, exogenous flows generate the low-frequency tilt, and several additional testable restrictions match well with the data. Notwithstanding, we conduct a formal test that offers a complementary perspective on endogeneity. Readers who subscribe to the above observations can skip the remainder of this section without any loss of continuity.

Our test to account for endogeneity is based on the “Granular Instrumental Variable” (GIV) approach proposed by Gabaix and Koijen (GK, 2022a,b).³⁰ Intuitively, this approach extracts large idiosyncratic shocks to fund flows and uses them as a source of exogenous variation. Indeed, idiosyncratic flow shocks emanating from individual funds are unlikely to be caused by factor returns (or by omitted variables correlated with those returns) since any such influence would systematically affect all funds.

Table 11: Quasi-Exogenous Flows

This table reports the regression results using granular instrumental variables (*GIV*, Gabaix and Koijen 2022a,b) for hedge- and mutual-fund flows. The estimation procedure is described in detail in Internet Appendix I6. *t*-statistics are calculated based on Newey-West standard errors with 13 lags. The sample period is 1994–2020.

	Model (1): Time Series		Model (2): Cross Section	
	(1)	(2)	(3)	(4)
	FREQ = LOW	FREQ = HIGH	FREQ = LOW	FREQ = HIGH
ZHF-FREQ	0.323 [5.57]	-0.007 [-0.19]	0.163 [3.21]	0.063 [1.48]
ZMF-FREQ	-0.600 [-8.22]	-0.179 [-3.42]	-0.493 [-3.38]	0.048 [1.12]
η_1 - Sect	0.411 [2.76]	-0.097 [-1.76]	0.042 [0.27]	-0.030 [-0.45]
η_2 - Sect	-0.150 [-1.76]	-0.018 [-0.23]	-0.031 [-0.27]	-0.062 [-0.69]
η_1 - Econ	0.010 [0.08]	0.032 [0.35]	0.462 [2.64]	-0.051 [-0.40]
η_2 - Econ	0.179 [1.96]	-0.047 [-0.67]	0.176 [1.76]	-0.027 [-0.34]
PC1 - Sect	0.173 [1.96]	0.066 [1.10]	0.343 [3.26]	0.011 [0.17]
PC2 - Sect	-0.101 [-1.14]	0.018 [0.43]	-0.045 [-0.40]	0.011 [0.20]
PC1 - Econ	-0.113 [-1.43]	0.036 [0.95]	-0.291 [-3.59]	0.084 [2.09]
PC2 - Econ	0.045 [0.72]	0.010 [0.26]	0.088 [1.21]	0.006 [0.15]
Controls	Yes	Yes	Yes	Yes
N	323	323	324	324
R ²	55.6%	38.4%	42.6%	34.2%

³⁰ To complement the GIV approach, we present in Internet Appendix I6 an additional test that exploits seasonal variations in fund flows motivated by fund clients’ cash needs unrelated to the stock market (Etula et al. (2020)). Specifically, we use abnormal December flows—defined as the difference between December flows and the average monthly flows over the preceding 11 months—as a source of exogenous variation in fund flows. For non-December months, abnormal December flows are set to zero. We estimate 2SLS regressions of decomposed returns on decomposed flows, instrumented by the low- and high-frequency components of abnormal December flows. These regressions yield results consistent with those based on GIVs, confirming the low-frequency tilt. However, these results should be interpreted with caution due to the small number of non-zero observations of abnormal December flows.

We estimate two versions of GIVs, exploiting either time series or cross-sectional variations. The procedure, which follows GK (2022a, Section 4.3) and is detailed in our Internet Appendix I6, can be summarized as follows. We first estimate idiosyncratic shocks from the residuals of either aggregate time series or fund-level panel regressions of flows. Then, we control for systematic factors affecting equity demand using holdings data. Specifically, we extract sector-specific factors from the holdings of institutions identified as hedge or mutual funds, and economy-side factors from the holdings of other institutions. Some factors are estimated accounting for observable characteristics of institutions (e.g., size and active share), while latent factors are estimated non-parametrically through PCA. Finally, we focus on large idiosyncratic shocks—large enough to influence factor returns—by weighing them by institutions’ equity shares.

The results, reported in Table 11, are in line with our main findings: the impact of flows on factor returns is tilted toward low frequency for both hedge and mutual funds, with opposite signs. The magnitude of the tilt is comparable with that reported in Table 5. All in all, the findings in this section lend support to a causal interpretation of the (frequency-dependent) effect of flows on factor returns.

Conclusion

We document that the *persistent* components of aggregate capital flows to hedge funds predict factor returns *out of sample*, with a power far stronger than extant predictors. Total flows and their transient components display no such OOS power, even though transient flows are more predictable than persistent flows. Persistent, but not transient, mutual fund flows also possess predictive power but it is weaker than hedge funds’.

To uncover the sources of this power, we decompose flows and factor returns into low- and high-frequency components through spectral analysis. We find that flows exhibit a *low-frequency tilt* in that they have a much larger effect on returns at low frequency than at high frequency. For hedge funds, this tilt accounts for the predictive power of persistent flows, and is primarily due to fund managers rather than fund investors. We address the possible endogeneity by employing the GIV approach.

We rationalize the low-frequency tilt, and reinforce the plausibility of a causal interpretation, by proposing a novel equilibrium *model* featuring rational, constrained, fund managers. Managers choose how aggressively to invest according to the persistence of capital supply in the arbitrage sector. Because of constraints on leverage, they aggressively deploy persistent flows in factor strategies, while parking transient

flows into liquid assets. The low-frequency tilt also produces factor momentum. We report empirical evidence supporting the model's mechanism and highlighting its broad implications.

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